NOT MEASUREMENT SENSITIVE

MIL-HDBK-775A
15 November 2021
SUPERSEDING
MIL-HDBK-775
19 April 1996

DEPARTMENT OF DEFENSE HANDBOOK

FOAM-IN-PLACE PACKAGING PROCEDURES FOR

This handbook is for guidance only.
Do not cite this document as a requirement.

AMSC N/A **AREA PACK**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

FOREWORD

1. This handbook is approved for use by all Departments and Agencies of the Department of Defense (DoD).

2. This handbook provides the requirements and procedures (techniques) for packaging supplies using foam-in -place (FIP) materials.

3. Comments, suggestions, or questions on this document should be addressed to Director, Army Sustainment Command, Packaging, Storage, and Containerization Center, ATTN: AMAS-SPI, Tobyhanna, PA 18466-5097 or emailed to usarmy.tyad.usamc.mbx.pt@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

CONTENTS

PARAGRAPH		PAGE
	FOREWORD	ii
1.	SCOPE	1
1.1	Scope	1
1.2	Classification	1
2.	APPLICABLE DOCUMENTS	1
2.1	General	1
2.2	Government documents	1
2.2.1	Specifications, standards, and handbooks	1
2.2.2	Other Government documents, drawings, and publications	2
2.3	Non-Government publications	2
2.4	Order of precedence	3
3.	DEFINITIONS	3
4.	GENERAL GUIDANCE	4
4.1	Materials	4
4.1.1	Foam chemicals	4
4.1.2	Toxic and hazardous materials	4
4.1.3	Use of recycled polyurethane	5
4.2	Sensitive items	5
4.3	Safety and health considerations (Occupational and environmental)	5
4.3.1	Purging dispensing head	6
4.3.2	Pyrolysis	6
4.3.3	Combustibility	6
4.3.4	Personal protective measures	6
4.3.5	First aid	6
4.3.6	Vapor detection	7
4.3.7	Decontamination	7
4.3.7.1	Clothing	7
4.3.7.2	Spills	7
4.3.7.3	Empty drum decontamination	7
4.3.8	Waste management	7
4.3.8.1	Purging solvents	8
4.3.8.2	Decontamination waste	8
4.3.8.3	Empty drum reclamation	8

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

CONTENTS

PARAGRAPH		PAGE
4.4	Unit protection	8
4.5	Foam dispensing equipment	8
4.6	Foaming conditions	8
4.7	Predetermination of foam thickness	8
4.8	Determination of foam volume (quantities)	9
4.9	Utilization of restraining devices	9
5.	DETAILED REQUIREMENTS	9
5.1	General	9
5.1.1	Technique I, split pack, standard	9
5.1.1.1	Technique I, procedure	9
5.1.2	Technique II, split pack, alternate	10
5.1.2.1	Technique II, procedure	11
5.1.3	Technique III, split pack, inverted	11
5.1.3.1	Technique III, procedure	12
5.1.4	Technique IV, foam-in-bag	12
5.1.4.1	Technique IV, procedure	12
5.1.5	Technique V, special technique modified	13
5.1.6	Technique VI, foamed container	13
5.1.6.1	Technique VI, procedure	14
5.1.7	Technique VII, encapsulated pack	14
5.1.7.1	Selection of applicable design procedure for technique VII	15
5.1.7.2	Technique VII procedure	15
5.1.7.3	Special marking for technique VII packs	17
5.1.7.4	Reusability of technique VII packs	17
5.1.8	Technique VIII, preformed molding	18
5.1.8.1	Technique VIII, procedure	18
5.2	Performance characteristics	19
5.2.1	Exterior appearance	19
5.2.2	Rough handling damage	19
5.2.3	Shock mitigation (flexible foams)	19
5.2.4	Pack opening	19
5.2.5	Completeness of fill	19
5.2.6	Foam adherence	20
5.2.7	Release of items	20

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

CONTENTS

PARAGRAPH		PAGE
5.2.8	Item condition	20
5.2.9	Conditioning of pack	20
5.3	Reusability	20
5.4	Marking for shipment and storage	20
5.4.1	Special marking and labeling	20
5.5	Foam workmanship	20
5.5.1	Rigid foams	20
5.5.2	Flexible and semirigid foams	21
6.	NOTES	21
6.1	Intended use	21
6.2	Ordering data	21
6.3	Sampling for Group B inspection	22
6.4	Subject term (keyword) listing	23
6.5	Changes from previous document	23

FIGURES		PAGE
1.	Design for flexible foam packs	24
2.	Design of 0.5 pound density semirigid foam packs	26
3.	Determination of tube/bag width for technique IV	28
4.	Split pack, standard	29
5.	Split pack, alternate	30
6.	Split pack, inverted	31
7.	Foam-in-bag	32
8.	Foam container	33
9.	Encapsulated pack (Design 1)	34
10.	Encapsulated pack (Design 2)	34
11.	Encapsulated pack (Design 3)	35
12.	Encapsulated pack (Design 4)	36
13.	Encapsulated pack, special marking locations	38
14.	Preformed mold	39

APPENDIXES

A.	Quality assurance provisions	40
B.	English-metric conversions	47

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

CONTENTS

TABLES

I.	Test required for the procedure and foam formulation utilized	43
II.	Standard English-metric equivalents	47
III.	MIL-HDBK-775 English-metric conversions	48

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1 Scope. This handbook provides the requirements and procedures (techniques) for packaging supplies using foam-in -place (FIP) materials. This handbook is for guidance only and cannot be cited as a requirement.

1.2 Classification. The procedures covered by this handbook should be of the following techniques, as specified (see 6.2).

Technique I - Split pack, standard
Technique II - Split pack, alternate
Technique III - Split pack, inverted
Technique IV - Foam-in-bag
Technique V - Special technique, modified
Technique VI - Foamed container
Technique VII - Encapsulated pack
Technique VIII - Preformed molding

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed below are not necessarily all of the documents referenced herein, but are those needed to understand the information provided by this handbook.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-203 - Paper, Kraft, Untreated
A-A-3174 - Plastic Sheet, Polyolefin
A-A-55057 - Panels, Wood/Wood Based; Construction and Decorative

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-117 - Bags, Heat-Sealable
MIL-PRF-121 - Barrier Materials, Greaseproof, Waterproof, Flexible, Heat-Sealable
MIL-PRF-81705 - Barrier Materials, Flexible, Electrostatic Discharge Protective, Heat-Sealable
MIL-PRF-83671 - Foam-In-Place Packaging Materials, General Specification for

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129 - Military Marking for Shipment and St
MIL-STD-1916 - DoD Preferred Methods for Acceptance of Product
MIL-STD-2073-1 - Standard Practice for Military Packaging

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this handbook to the extent specified herein.

Code of Federal Regulations (CFR)

Title 40, Part 261 – Identification and Listing of Hazardous Waste

(Copies of this document are available online at https://www.ecfr.gov/.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Voluntary Product Standard PS 1 - Plywood, Construction and Industrial.

(Copies of this document are available online at https://www.apawood.org, or from the American Plywood Association, 7011 S. 19th Street, Tacoma, WA 98466-5333.)

ASTM INTERNATIONAL

ASTM D3953 - Standard Specification for Strapping, Flat Steel and Seals
ASTM D4727/D4727M - Standard Specification for Corrugated and Solid Fiberboard Sheet Stock (Container Grade) and Cut Shapes
ASTM D5330/D5330M - Standard Specification for Pressure-Sensitive Tape for Packaging, Filament Reinforced
ASTM D5486/D5486M - Standard Specification for Pressure-Sensitive Tape for Packaging, Box Closure, and Sealing
ASTM D6123/D6123M - Standard Specification for Pressure-Sensitive Tape for Light-Duty Packaging and General Purpose Masking
ASTM D6199 - Standard Practice for Quality of Wood Members of Containers and Pallets
ASTM F1667 - Standard Specification for Driven Fasteners: Nails, Spikes, and Staples

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

(Copies of ASTM International standards are available at https://www.astm.org or from ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959, telephone (877) 909-2786 (USA and Canada), (610) 832-9585 (International), or email: service@astm.org.)

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of ANSI standards are available at https://www.asq.org.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

This section is not applicable to the handbook.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

4. GENERAL GUIDANCE

4.1 Materials. Associated materials required to facilitate the development of FIP packs should be as specified herein and up-to-date safety data sheets (SDS) should be included.

4.1.1 Foam chemicals. The FIP chemicals and characteristic end product should conform to the requirements of MIL-PRF-83671 and should be selected for the prescribed procedure from the approved classes and grades/categories specified (see 6.2). Additionally, the applied foams should conform to the workmanship requirements of 5.5.

4.1.2 Toxic and hazardous materials. The individual foam chemicals are classified as toxic or hazardous in their original state and, also when combined during the mixing process. Once reacted, the foam is hazardous only as dust or if ingested. Further hazards can occur when materials are not stored properly, dispensing equipment is not maintained correctly, or purging operations are improperly performed (see 4.3.1). Additional information on safety and maintenance procedures for the storing, handling, and use of the involved chemicals and equipment can be found in the respective manufacturer's safety and maintenance manuals, the availability of which is required by the Occupational Safety and Health Administration (OSHA) (see 4.3.). It should be noted, too, that where available, ozone-safe chemical systems (those combinations of chemicals and dispensing equipment that do not employ chlorofluoro carbon-blowing agents) should be used. A partial list of significant foam chemicals along with their hazards and potential effects are given below.

a. Di and polyisocyanates (component A). This is a skin irritant, harmful to eye tissues, that causes bronchospasm and respiratory tract sensitization in susceptible individuals.

b. Trichlorofluoromenthane (typical blowing agent). Prolonged breathing of high concentration can cause unconsciousness and anesthesia leading to asphyxiation.

c. Silicone oils (cell control agents). Low toxicity but repeated contact should be avoided.

d. Phosphate esters (for flame retardancy). Phosphate esters are readily absorbed through skin causing blurred vision, nausea, headaches, abdominal cramps, coughing, and breathing difficulties.

e. Amines (catalysts). Vapors are eye irritants and can irritate mucous membranes. Aliphatic amines can produce skin burns and dermatitis and are a respiratory irritant.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

f. Methylene chloride (purging solvent). Methylene chloride is a skin and respiratory irritant that can also act as a chemical asphyxiant. It is a solvent that can result in skin, eye, liver, respiratory tract, and central nervous system toxicity.

g. Ethylene glycol monothyl ester (cleaning solvent). It is a skin and respiratory irritant.

h. Dioctyl or diallyl phthalate (plasticizer/lubricant). It is a skin and respiratory irritant.

4.1.3 Use of recycled polyurethane. Recycled polyurethane should be the same category, class, or grade of MIL-PRF-83671 as that of the new foam to be dispensed. The use of recycled polyurethane should be limited only to the extent that formed foam packs meet the performance requirements of this standard and that such reused foam does not exceed 50 percent of the total foam volume. Recycled polyurethane foam should not be used as filler for technique VII packs.

4.2 Sensitive items. Unless the contract specifies otherwise (see 6.2), FIP techniques should be used only on items that are not susceptible to either electrostatic discharge (ESD) or high temperatures because FIP operations are enormous generators of both ESD voltage and exothermic heat (up to 150° F.) (66° C.). When the use of FIP techniques are permitted by contract on ESD sensitive items, they should be enclosed in a static-shielding wrap or barrier material in accordance with MIL-PRF-81705. Parting or separating film should be limited to electrically conductive, anti-static types also in accordance with MIL-PRF-81705. When a FIP technique is permitted by contract for use with an item that is sensitive to high exothermic heat, the item should not be placed in direct contact with the separating film or rising foam. The preferred FIP method for packing such heat-sensitive items is technique VIII, preformed molding. Note: Exothermic heat is not a concern for this technique.

4.3 Safety and health considerations (occupational and environmental). All chemicals and materials, required for FIP packaging should be ordered and furnished with up-to-date safety data sheets. For contract application, the appropriate OSHA standard should be specified and followed. Any potential hazards resulting from the use of FIP procedures should be minimized through the use of proper preventative measures and controls. When acquiring FIP services from outside contractors, specific safety guidance can be obtained from specific manufacturer's required safety and maintenance manuals. Both contractors and Government personnel can also obtain such safety guidance from local industrial hygiene, medical, and safety personnel. Moreover, any FIP dispensing operations within DoD installations and facilities (land and afloat) should have medical service available for environmental health or safety approval regarding ventilation and protective equipment requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

4.3.1 Purging dispensing head. If solvent purging of the dispensing head is used, it is recommended that the discharge solvent be directed into a container containing approximately 1 inch of water. This action will limit the escape of solvent (methylene chloride) vapors (see 4.1.2). Workers should use the appropriate skin, eye, and respiratory protection as designated by local industrial hygiene personnel.

4.3.2 Pyrolysis. Smoking in foaming areas should be prohibited. This is required due to the pyrolysis breakdown products and not the flammability characteristics of the foam chemicals.

4.3.3 Combustibility. Combustion of solid foam and the foaming ingredients produce toxic gases, as do most packaging materials. In the event of fire, the local fire departments should be immediately notified. Full face, self-contained breathing apparatus should be employed for these fires. For small fires, CO2 or dry chemical fire extinguishers are suitable.

4.3.4 Personal protective measures. Protective measures involved with foaming operations include proper training, assignment of personal safety equipment, location of approved emergency facilities, and the designation of safe operating and reporting procedures. Eye wash fountains and showers should be located in the immediate foaming area or readily accessible thereto.

4.3.5 First aid. If chemicals containing isocyanates (primarily component A) are splashed on the skin, treatment should be started immediately as follows:

a. Shower immediately (major splash only).

b. Apply tincture of green soap (full strength) or rubbing alcohol to the remaining contaminated area.

c. After 2 to 4 minutes, wash off the tincture of green soap or rubbing alcohol with water. If there is still an indication of isocyanate present, another application will be necessary. If the isocyanate contamination is on the face, care should be taken to prevent getting the tincture of green soap in the eyes.

d. When polyol component (component B) is splashed on the skin, it should be washed away with soap and water.

e. If either component is splashed in the eyes, the eyes should be flushed immediately with clean water for 15 minutes. The eyelids should be held apart during this treatment. Medical attention should be promptly obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

f. If a person accidentally swallows the isocyanate component and is conscious, dilute by having the victim drink milk or water and transport him to a physician immediately. If a person is overexposed to the vapors of components, he should be removed from exposure immediately and transported to a physician. Oxygen may be administered by qualified personnel en route if the exposed person has difficulty breathing.

4.3.6 Vapor detection. Field detection must be performed under the direction of trained industrial hygiene personnel. Interpretation of sampling data must be performed by appropriate industrial hygiene or medical personnel. The maximum allowable concentration of disphenyl methane diisocyanate in a worker's breathing zone is 0.005ppm.

4.3.7 Decontamination.

4.3.7.1 Clothing. Clothing splashed with the polyol component should be removed and washed in an aqueous detergent solution before wearing again. Clothing splashed with the isocyanate component should be removed immediately, the isocyanate scraped off the clothing, and the clothing washed in an aqueous detergent solution. In the case of large spills that have soaked through clothing, wash clothing in an aqueous solution before discarding.

4.3.7.2 Spills. Accidental floor spills may be decontaminated by covering with an absorbent material such as absorbent shredded paper, sawdust, Fuller's earth, or other absorbent materials. If amine fumes are present, a vapor cartridge-type respirator should be used. Pour liquid decontaminant consisting of 90 to 94 percent water, 4 to 8 percent concentrated ammonium hydroxide, and 2 percent detergent on the spillage and allow to react for at least 10 minutes. Carefully remove all residues from the spill, placing them in open containers, and add further amounts of liquid deconcentrate. After removing containers to a safe place, the floor may be washed with liquid decontaminate, soapy water, or an ammonia solution.

4.3.7.3 Empty drum decontamination. Polyol drums may be decontaminated with soapy water and reclaimed through any suitable outlet. Isocyanate drums may be decontaminated by slowly adding the liquid decontaminant. Under no circumstances should the drums be closed. After 24 to 48 hours, the drums may be drained (see 4.3.8) and pierced to prevent reuse.

4.3.8 Waste management. All waste (liquid or solid foam) should be disposed of in a manner that will minimize pollution and enhance the environment. The first consideration in waste management should be reuse or recycling. Applicable directives and local environmental authorities should be consulted before complete disposal of these materials is attempted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

4.3.8.1 Purging solvents. Some operations use solvents such as methylene chloride (dichloromethane) to clean foam mixing and dispensing equipment. Wastes from these mixtures of solvent and foam components must be managed as a hazardous waste with Environmental Protection Agency No. F001 (CFR 40, 261).

4.3.8.2 Decontamination waste. The appropriate regulatory activity designates acceptable methods of disposing of reacted (neutralized) waste solution.

4.3.8.3 Empty drum reclamation. Reclamation of decontaminated drums is preferable to landfill disposal. Alternatively, empty drums, as defined in 40 CFR, 261.7, may be accepted at sanitary landfills by special arrangement and with the concurrence of state regulatory authorities.

4.4 Unit protection. Unit protection requirements should be applied in accordance with MIL-STD-2073-1 or other applicable documents as specified (see 6.2) prior to the application of the specified FIP technique. When required, wrap-type cushioning material should be used to cover projections, sharp edges, or curves of the item.

4.5 Foam dispensing equipment. Any equipment may be used provided that as a foaming system consists of the chemicals and the dispensing equipment together. Both must be compatible and produce finished foam that is in accordance within requirements of MIL-PRF-83671.

4.6 Foaming Conditions. All components, surfaces, wraps, and void areas of the proposed foam pack to be filled should be conditioned and maintained at a constant temperature within the limiting temperature range of 60° F. (16° C.) to 100° F. (38°C.) to ensure efficient, continuous foam production and for retention of optimum foam properties. Chemical ingredients should not be exposed to temperatures other than those recommended by the formulation supplier. Surfaces expected to come in contact with the foaming chemicals should be free of grease, oil, loose particles, moisture, and other deleterious (harmful) foreign matter.

4.7 Predetermination of foam thicknesses. Unless otherwise specified (see 6.2), minimum foam thickness determination should be made, as specified herein. In the case of flexible foams, cushioning requirements are based on item static-bearing stress and fragility (see figure 1). For rigid foams, minimum thickness should be 2 inches (5.08 cm). Where semirigid foam is intended for use with items weighing 150 pounds (68.10 kg) or less, constraints are also based on load bearing of the prepared item. In this specific application, a “new” bearing surface can essentially be formed around items of irregular configurations so that minimum allowable surface areas can be met (see figure 2 for 0.5 pound density). For special application of technique IV procedures, thickness determination listed in figure 3 will be used. However, the final thickness determination will be based on results of the rough handling tests specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

4.8 Determination of foam volume (quantities). The amounts of foam may vary based on the characteristic of the void area. In general, cool surfaces or low environmental temperatures will retard normal foaming action and require additional chemical components. Adjustment of foam quantities will be necessary after foaming initial packs.

4.9 Utilization of restraining devices. Foaming bucks or restraining blocks should be fabricated of any appropriate material in sufficient quantity to prevent distortion of container walls during foaming operations. The selected design should be based on the procedure technique used and of such construction as to allow for some venting and subsequent release of entrapped gases. Additional restraining devices should be used, as necessary, to limit the tendency for certain items to "float" or move from their designated orientation during foam rise. Pressures actually developed are dependent upon the class and grade of foam material used.

5. DETAILED REQUIREMENTS

5.1 General. This section contains the detailed, step-by-step requirements to be followed for accomplishment of each of the eight FIP packaging techniques classified in section 1. Each technique should utilize foam materials as specified in 4.1. All items to be packed using FIP procedures should be protected as specified in 4.4. Also, in each case, the term "prepared item dimensions" refers to the dimensions of the item along with all the pads, wraps, or other materials needed to prepare or make this item ready for the foaming operations. Unless otherwise specified within the requirements of a specific technique, the prepared items should be enclosed in a container as specified by the contract (see 6.2). Specific applications, requirements, or anticipated results that are unique to a particular technique preface each procedure.

5.1.1 Technique I, split pack, standard. This procedure is adaptable for many various items. It should be achieved using a minimum of two consecutive pours with time allowed between each pour to permit the foam to set (become tack free). The foam materials should be of the class specified (see 6.2).

5.1.1.1 Technique I procedure (see figure 4).

Step a. Select the best orientation for the item considering the ready availability of supporting surfaces. The orientation selected should also depend on the locations of critical item projections and the item void areas that could later cause removal of the item to be difficult.

Step b. Select container of appropriate dimensions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

Step c. Drape plastic sheet films as specified in A-A-3174 loosely inside container covering bottom and extending over flaps, where applicable. Smooth the sheet toward all corners. Tape, if necessary, to hold temporarily. The following formulas should be used to tabulate film requirements:

Sheet length = 2 x depth + 1.5 x length of container interior
Sheet width = 2.5 x depth + width of container interior

Step d. Place prefoamed support block to support weight of prepared item to required height in container bottom. Position to support item uniformly.

Step e. Remove item. Then dispense enough foam mix into the container to encase block(s) and allow to rise to the required height, which is approximately to the middle of the item. This is the first pour.

Step f. Fold plastic film inward completely covering foam before it completes its rise.

Step g. Immediately place and hold prepared item on top of the plastic film/foam and the foam and the foam support until foam has risen completely and set enough to support the item. (Some items have a tendency to float during foaming.)

Step h. Drape a second sheet of plastic film loosely inside the container, covering the prepared item and foam. Extend film over flaps. Tape film, if necessary, to hold temporarily.

Step i. Dispense a sufficient amount of foam mix into the container to surround the item and fill the container with expanded foam. This is the second pour.

Step j. Fold the plastic film inward when certain that the rising foam will fill the container. Fold the container flaps over, but do not secure. If the container becomes overfilled, trim off excess.

Step k. Apply appropriate closure. Allow time for foam to set before moving container.

5.1.2 Technique II, split pack, alternate. This procedure is a variation of technique I, split pack, that provides a more simple means of completing foam packs for use in remote areas or aboard ship and for retrograde materials or items that must be rapidly deployed. Time should be allowed between each pour to permit the foam to set (tack free) properly. Note that prefoamed support blocks are not used. Foam materials should be of the class specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

5.1.2.1 Technique II procedure (see figure 5).

Step a. Select the best orientation for the item considering the ready availability of support surfaces. The orientation selected should also depend on locations of critical items projections and the item void areas that may later cause difficult removal of the item. Protective material should be used only as required to prevent puncture of the plastic sheet which would subsequently alleviate the danger of the poured foam contacting the item or locking the molded halves together.

Step b. Select specified container to hold prepared item.

Step c. Drape sheet plastic film as specified in A-A-3174 loosely inside the container covering bottom and extending over flaps. Tape, if necessary, to hold temporarily.

Step d. Dispense sufficient amount of foam mix into the container to center item in container. Fold plastic film inward completely covering foam as it completes its rise.

Step e. Place prepared item, overwrapped in plastic (A-A-3174), on the foam surface after foam has risen and set enough to support the items. (The item should be centered in container.)

Step f. Drape a second sheet of plastic film loosely inside the container covering the prepared item and foam. Extend film over flaps. Tape film, if necessary, to hold temporarily.

Step g. Dispense a sufficient amount of foam mix into the container to surround the item and fill the container with expanded foam.

Step h. Fold the plastic film inward when certain that the rising foam will fill the container. Fold container flaps over, but do not secure. If the container becomes overfilled, trim off excess.

Step i. Apply applicable closure. Allow time for foam to set before moving container.

5.1.3 Technique III, split pack, inverted. The inverted pack should not be used for items that must always remain in an upright position to prevent internal damage. Foam material should be of the class specified (see 6.2)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

5.1.3.1 Technique III procedure (see figure 6).

Step a. Select the best orientation for the item considering the ready availability of supporting surfaces. The orientation selected should also depend on locations of critical item projections and the item void areas that may later cause difficult removal of the item.

Step b. Select specified container to hold prepared item.

Step c. Cut a pad(s) of foam the same length and width as the container's inside dimensions and a depth equal to the thickness of the foam to be poured. This will result in the item being centered vertically in the container.

Step d. Place prepared item upside down on the foam pad, centered in the container.

Step e. Drape a sheet of plastic film as specified in A-A-3174 loosely inside the container covering the prepared item and foam pad and extending over the flaps. Tape, if necessary, to hold temporarily.

Step f. Dispense enough foam mix into the container to surround the item and fill container with expanded foam.

Step g. Fold the plastic film inward when certain that the rising foam will fill the container. Fold container flaps over, but do not secure. If the container becomes over filled, trim off excess before taping.

Step h. Foam must be able to support item weight before container is turned upright. After foam is cured 15 minutes, invert container to upright position.

5.1.4 Technique IV, foam-in-bag. This procedure is adaptable to a wide variety of items regardless of dimension. When items exceeding approximately 32 inches in length are being foamed, it is preferable to use a multiple bag unit design. Foam material should be of the class specified (see 6.2).

5.1.4.1 Technique IV, procedure (see figure 7).

Step a. Select the best orientation for the item while considering the ready availability of supporting surfaces. The orientation selected should also depend on locations of critical item projections and the item void areas that may later cause difficult removal of the item.

Step b. Select container of appropriate dimension (specific container or restraining device) based on prepared item dimension.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

Step c. Determine dimensions (see figure 3) and fabricate or select bags (or tubes) fabricated from plastic film as specified in A-A-3174 in the width (diameter) and thickness, as specified (see 6.2), that are necessary to hold the item immobile in the container. Tubes should be heat sealed transversely at the center line to form two compartments. The flat tubing dimension should be determined by the interior dimension of the container or restraining device. Formulas should be used as follows:

Tubing length (LT) = 2 x depth + width + 4 inches (10.16 cm)
Tubing width (WT) = length + width + 2 inches (5.08 cm)

Step d. Prepare a sufficient number of cut prefoamed blocks (see 4.1.3) to support the prepared item to its required height above the bottom of the container during the foaming operation. Place these prefoamed blocks inside the bags or tubes in close proximity to the common heat seal; then position inside the container. The bags or tubes may be taped or clipped to the container walls or flaps to hold them in place to facilitate easier dispensing of foam mix.

Step e. Position the prepared item in the center of the container supported by the foam blocks inside the tubes.

Step f. Dispense equal amounts of the foam mixture into each set of bags or tubes. Hold the prepared item in place until foam has risen and set enough to support the item. When foam reaches half of its rise, remove any tape or clips holding the bags to the container.

Step g. Overlap the tube or bag ends, and fold the container flaps over temporarily until rise has stopped. Do not secure. Open the container to make sure that the item is immobilized by the bags and allow the foam to cure 15 minutes before handling. Close flaps and secure with tape.

5.1.5 Technique V, special technique, modified. Whenever the particular nature of an item (configuration, size, fragility, or any other unique aspect or combination thereof) dictates deviations from the FIP procedures specified in techniques I through IV, those deviations must be recognized as special, the necessary packaging adaptations should be invoked, and the resultant procedure should be designated, technique V. When employing this technique in the acquisition process, the special procedures, instructions, and descriptions necessary should be included as shown in 6.2; as a procedure by Government personnel, apply as specified herein (see 1.2)

5.1.6 Technique VI, foamed container. This procedure is essentially a split pack using special fabrication techniques and should be used primarily with large, heavy items. Foam material should be of the class specified (see 6.2). Specific fabrication procedures

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

should be developed by the fabricator to meet the requirements of the engineering drawing/specification (see 6.2.). The following design steps are intended for general guidance only.

5.1.6.1. Technique VI procedure (see figure 8).

Step a. Set up the specified mold for fabricating the foamed container (a simulated item can be used). Mold design should allow easy disassembly that will ensure against damage of the foamed container upon removal.

Step b. Apply release material (plastic sheet (A-A-3174), wax coatings, etc.) to the inside surfaces of the mold and completely around or on the surfaces of the item. Tape plastic sheet, if necessary, to hold in place temporarily.

Step c. Position inserts such as base with skids, push plates, tie down rings, or reinforcements in the mold and secure.

Step d. Position and secure items in the proper location in the mold. The item can be suspended from a hoist or supported with prefoamed blocks.

Step e. Dispense the required quantity of foam into the mold so that the total rise reaches approximately one-half of the depth of the item. Where consecutive pours are necessary, time is required between each pour to allow the foam to set properly.

Step f. Apply release materials to the top surfaces of the risen foam.

Step g. Dispense the required quantity of foam into the remaining void so that it surrounds the prepared item completely. Consecutive pours are performed as in step e above.

Step h. When it can be seen that the rising foam will fill the mold, place and secure the lid. The lid may contain holes to allow release of excess expanded foam and gases. Allow the foam to cure for 15 minutes before moving or opening the mold.

Step i. Remove foamed container from mold. Apply edge protectors for strapping.

Step j. Coat outside surfaces of foamed container as required by engineering drawing/specifications. Assemble and secure closure.

5.1.7 Technique VII, encapsulated pack. In this procedure, items are completely surrounded (encapsulated) in a minimum 2-inch thickness of foam to achieve water vapor protection. Foam material should be of the class specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

5.1.7.1 Selection of applicable design procedure for Technique VII (see 6.2):

Design 1. Small items, 0 to 100 pounds (45.4 kg) (see figure 9).

Design 2. Intermediate items, 75 (34.05 kg) to 125 pounds (56.75 kg) (see figure 10).

Design 3. Intermediate items, 100 (45.4 kg) to 250 pounds (113.50 kg) (see figure 11)

Design 4. Large or dense items, 150 (68.10 kg) to 400 pounds (181.6 kg) see (figure 12).

Design 5. Special items, as specified by procuring activity.

5.1.7.2 Technique VII procedure.

Step a. Select the best orientation for the item considering the ready availability of supporting surfaces. The orientation selected should also depend on locations of critical item projections and the void areas that may later cause removal of the item to be difficult.

Step b. Select container and applicable designs based on item dimension, weight, minimum foam thickness (see step g below), wrap, protective sleeve, and the orientation determined in step a above. For design 2, the container should be fastened to the skid by driving nails in two parallel rows, spaced alternately 3 inches (7.62 cm) apart, through the batten and container into each skid post. For designs 3 and 4, first the plywood base should be secured to the skid. This is done by driving nails in two parallel rows, spaced alternately 3 inches (7.62 cm) apart, through the base as shown in figure 11 and 12. Also, for design 4, the top frame should be constructed prior to securing it to the container shell and then pressed into position and secured to the shell using metal staples or nails in the same way that the shell is secured to the base. This must be accomplished prior to the final pour of foam.

(1) For securing bases and containers to skids, use pallet nails conforming to ASTM F1667, minimum of 2.5 inches (6.35 cm) long.

(2) All wood used in type VII packs should conform to ASTM D6199, groups II or III.

(3) All plywood bases used should conform to A-A-55057, Type A, PS-1 full exterior bond classification and a minimum grade C-C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

(4) Unless otherwise specified (see 6.2), containers should be fabricated from fiberboard conforming to ASTM D4727 grade V3c, type CF, class WR. Container dimensions should be inside measurements and should be cited in the sequence of length, width, and depth. The length dimensions should be the larger of the two dimensions of the open face of the box, and the depth dimension should be perpendicular to the length and width.

Step c. Prepare properly sized prefoamed blocks to support the item in a stable position during the foaming procedure. Blocks must be cut and positioned so that the direction of foam rise (maximum compressive strength) is supporting the item. This is very important on heavy items to prevent blocks from collapsing. Class and grade of foam should be identical to the foam ingredients to be used (see 4.1.1). The thickness of the blocks should be determined by the requirement for critical centering of the item between the top and bottom planes of the receptacle. Additionally, load-bearing material such as fiberboard or plywood may be further required to distribute concentrated loads over the area of the foam blocks thereby preventing compression and damage to the foam blocks prior to foaming.

Step d. With the item supported on the blocks, determine the cut line in the horizontal plane that will best permit removal of the top cut section of the foamed pack and subsequent removal of the item from the remaining bottom section.

Step e. A protective sleeve should be fabricated to contour to the item as closely as possible and be positioned horizontally around the item so that the cut line marking to be applied on the exterior of the container will be located at the vertical center of the sleeve. The sleeve should be fabricated and formed using fiberboard material conforming to type CF, grade V3c, and class WR, ASTM D4727. The sleeve should be secured in place with tape conforming to ASTM D5330 and drawn tightly to the item under adequate tension. This sleeve is designed to provide protection to the item during the cutting operation when opening, and to facilitate the removal of both the top of the container and the item from the base section. For large container designs, at this point, preliminary foaming into selected plastic bag sizes conforming to MIL-DTL-117 should be used for filling large void areas between the sleeve and the item (regularize configuration). Similarly, properly sized filler blocks built up of fiberboard sheet stock may be used. This process will ensure support to the sleeve against anticipated external pressure from the expanding foam in step g below. Care should be exercised to ensure that there are no projections beyond the plane formed by the sleeve.

Step f. Two pieces of barrier material, one conforming to MIL-PRF-121, Type I (medium duty) and the second conforming to A-A-203, 40-pound (18.16 kg) basis weight (minimum) should be prepared of such dimension as to completely encircle the item with sleeve attached. The first wrap, MIL-PRF-121, should be applied with the “face” or wax free side toward the item. Both wraps should be applied in such a manner to conform to the contour of the item and should not be loosely applied. Caution should be taken to

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

ensure that no rips, tears, or holes are present in the wraps exposing any parts of the item prior to application of foam. The overlaps (wrap joints) should be rolled and pressed to ensure that the foam does not enter the wraps and contact the item. Secure and seal outer Kraft wrap joints, tucks, folds, seams, etc. with ASTM D6123 tape, 2-inch (5.08) width minimum.

Step g. The wrapped item should have a minimum clearance between the container walls and the outermost projections of the wrapped item, as specified (see 6.2).

Step h. The amount of foam specified (see 6.2) should only be used as a guide. Exact amount required is the responsibility of the packaging activity. Dispense foam into the void using successive pours, as required, ensuring that any additional pour is dispensed at the tack free point of the previous pour. The amount actually used per pack can be greatly influenced by the type of equipment used, foam formulation, atmospheric conditions, facilities, etc. Sufficient quantities of foam should be dispensed to provide total volume fill. Furthermore, for design 4, the polyurethane foam must lay level with the top of the 2- by 2-inch (5.08 cm) top frame around the circumference of the pack to permit the plywood cover to be nailed flush to that 2- by 2-inch (5.08 cm) top frame and still meet the performance characteristics of the pack. Also for design 4, the scored lip, 1.5-inch dimension (3.18 cm) of the shell, should be folded over 90° onto the top frame and secured in place (see figure 12, inset).

Step i. Close container with ASTM D5486, 2-inch (5.08 cm) wide tape for designs 1, 2, and 3. Design 4 should be closed by nailing a minimum .375-inch (0.95 cm) thick plywood cover to the top frame using pallet nails conforming to ASTM F1667, minimum of 1.625 inches (4.13 cm) long. Nailing should be in a staggered pattern with nails not over 4 inches apart. The plywood cover should not be nailed to the top frame until the foam pack has been inspected for 100 percent fill and allowed to set up for a minimum of .12 hours. All plywood covers should conform to A-A-55057, Type A, PS-1 full exterior bond classification and a minimum grade C-C.

5.1.7.3 Special marking for technique VII packs. Each design in technique VII should include the special markings as illustrated in figure 13. (Special markings should be limited to stenciling, printing, or silk screening per MIL-STD-129. Labels are not permitted for any marking other than bar coding and shipping label.)

5.1.7.4 Reusability of technique VII packs.

a. Reusability of technique VII packs is limited to the return of the unserviceable part.

b. Once the pack is opened, the item is subject to environmental deterioration and should be stored indoors under controlled conditions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

c. All technique VII packs opened for quality conformance inspection should not be reused or repaired and should be disposed of.

5.1.8 Technique VIII, preformed molding. This procedure is specifically intended to form individual mold halves for items using a specially designed molding box or jig. An essential part of this procedure is that a very rigid molding fixture is required. It is also the means for vacuum withdrawal of air to ensure close conformity of the plastic release material to the simulated surfaces of the item, the container, and the mold mating surfaces. Technique VIII readily permits the forming of foam molds that would fit a variety of shipping containers other than those having a solid, rectangular shape. Thus, contoured molds may be formed intended for shipment in cylindrical containers or cans. Mating molds may be fabricated in advance of the actual packaging process. Due to the nature of the performed molding procedure, a single pour of foam is preferred for each mold half. Foam material should be of the class specified (see 6.2).

5.1.8.1 Technique VIII procedure (see figure 14).

Step a. Select applicable prefabricated mold fixture.

Step b. With the vacuum source connected, drape a 2-mil polyethylene sheet over the simulated item half and fixture flat surfaces, allowing sufficient overlap to completely cover the underside of the securing lid. With the vacuum source operating, ensure that the plastic sheet is pulled flat and tightly creased, when necessary, over these surfaces. Leave the vacuum source on during the following steps.

Step c. Dispense sufficient foam (single pour) into the prepared mold fixture.

Step d. Secure the plastic-lined lid over the rising foam in the fixture. Allow the filled mold to set for 15 minutes without handling or until it has sufficiently cured to prevent distortion of the molded foam.

Step e. Carefully remove the finished mold half from the fixture and check for conformity on all surfaces.

Step f. Duplicate steps b through e to produce a mating mold half.

Step g. Wrap the item, if required, in accordance with the configuration used for item shape simulation, as in step a. above. Place in formed mold halves and insert entire assembly in selected shipping container.

Step h. Close container and secure with tape.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

5.2 Performance characteristics. The packs formed using the FIP techniques specified in 5.1 should be acceptable for their intended purpose and should be capable of passing the tests and examinations specified in appendix A.

5.2.1 Exterior appearance. When inspected, as specified in appendix A, A.3.5.2, the exterior of the pack should be free from surface distortions measuring in excess of 0.25 inch (.64 cm) per 12 linear inches (30.48 cm) of container. Packs possessing skids or specialized bases and exceeding 150 pounds (68.1 kg) in gross weight should be checked for base distortion while the base is resting on a flat surface. Surface distortion measurements should not be more than 0.50 inch (1.27 cm) above the supporting surface at any point on the perimeter of the pack base. Opening and reusability instruction markings should be as specified for the selected procedural design and as further identified in 5.1.7.3.

5.2.2 Rough handling damage. There should be no evidence of exterior pack deficiency or item looseness within the foam molds that affects the integrity of the pack or damage to the item when subjected to the appropriate rough handling test specified in appendix A, A.3.5.3.

5.2.3 Shock mitigation (flexible foams). When specified (see 6.2), pack should be capable of protecting items against the imposition of accidental shocks above prescribed fragility levels. Shock mitigation afforded by the flexible FIP material should be consistent over the temperature extremes from -20° F. (-29° C.) to +125° F. (52° C.), and should not exceed the specified fragility level when subjected to the free fall drop test specified in A.3.5.3.1 of appendix A. There should be no exception to this requirement due to either weight or dimension (see appendix A, A.3.5.4).

5.2.4 Pack opening. FIP packs should be fabricated in such a manner that the container can be opened and reclosed without damage to the item or cushioning material. Failure to meet this requirement should be cause for rejection (see appendix A, A.3.5.4 and A.3.5.5). This is not the case with technique VII packs that have been opened for quality conformance inspection. Those should not be reused or repaired except for the return of an unserviceable part (see 5.1.7.4 and appendix A, A.3.5.8).

5.2.5 Completeness of fill. The resultant foam formed in the packs should provide, as nearly as possible, a complete fill of the intended void space. Severe rounding off of interior container corners, excessive noncontact (bridging) over desirable item bearing surfaces, large void areas, and evidence of incomplete bonding of prefoamed materials should not be permitted. For technique VII, foam must fill 100 percent of the container. No bridging (noncontact) over item surfaces should be permitted. Voids or evidence of incomplete bonding should not be permitted (see appendix A, A.3.5.4 and A.3.5.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

5.2.6 Foam adherence. There should be no evidence of foam breakthrough or penetration or adherence either on the item, intimate wraps (except special wraps) (see 4.2), sealed bags, inner surfaces of containers (except where foam is applied directly against container walls), or restraining bucks (see appendix A, A.3.5.4 and A.3.5.5).

5.2.7 Release of items. Items should be easily removed from the foam molds with no evidence of accidental sealing between wraps, parting films, or bags. There should be no evidence of locked items caused by improper application of multiple pours or excessive pressures created by void overfills (see appendix A, A.3.5.4 and A.3.5.5.)

5.2.8 Item condition. Item should show no evidence of damage resulting from the application of FIP materials. Apparent damage may include broken or distorted projections, loosening of mountings, broken wires, or scorched painted surfaces (see appendix A, A.3.5.4 and A.3.5.5). Humidity indicators, if used, should indicate a safe condition.

5.2.9 Conditioning of pack. When specified (see 6.2), the pack should be conditioned for 24 hours at a temperature of -20° F. (-29° C.) (see appendix A, A.3.5.4).

5.3 Reusability. Formed foam molds are reusable when the prescribed procedures are followed, as required in 5.1 (see appendix A, A.3.5.8), except for technique VII which does not provide water vapor protection after the original seal is broken and, thus should not be reused or repaired. Technique VII molds may be used to return an unserviceable part.

5.4 Marking for shipment and storage. All marking, including hazardous materials marking, should be in accordance with MIL-STD-129, with other special markings added, as prescribed herein (see appendix A, A.3.5.6)

5.4.1 Special marking and labeling. Marking or labeling instructions specified for opening and reusability (see 6.2) should be provided on the container (see appendix A, A.3.5.5).

5.5 Foam workmanship. The pack foam conforming to 4.1 and produced by the equipment specified in 4.5 should form a unicellular plastic foam suitable for the intended techniques listed in 1.2 (see appendix A, A.3.5.7).

5.5.1 Rigid foams. The foam pack formed by a FIP technique should be essentially homogeneous throughout with a uniform cell structure. There should be no splits, sparklers, void openings, or pockets over 0.50 inch (1.27 cm) in any direction that might have resulted from rapid formation or release of the blowing agent before the polymer structure reached sufficient strength to contain the gas. There should be no evidence of shrinkage such as would be apparent by wrinkles or indentations in the foam surface. Further, there should be no unexpanded resin, occlusions, or foam scorching as

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

evidenced by discoloration from the generation of excessive heat during the exothermic reaction. The fusion line between successive pours, unless separated by parting films, should be well knitted and should show no occlusions, cracks, or separations. The foam formed should not be soft, tacky, or brittle after curing.

5.5.2 Flexible and semirigid foams. The foam pack formed by the FIP process should not be soft, doughy, non-resilient, or brittle after curing. There should be no multiple holes, voids, or "blowouts" exceeding 2 inches (5.08 cm) in diameter. Contaminated, recycled foam or more than 50 percent recycled foam should not be integrated with the new foam pour in a completed pack. The cured foam shrinkage should not exceed 0.5 inch (1.27 cm) from any face of container inner walls.

6. NOTES

6.1 Intended use. This handbook is intended for use whenever guidance is needed to accomplish FIP packages of the type provided for in the eight techniques described herein and when duly authorized for use by the Government. It is intended for use in conjunction with MIL-PRF-83671 that specifies flame-retardant polyurethane FIP materials and applicable dispensing equipment for foam. The packs specified here-in may be fabricated by Government personnel, or the service may be acquired from contractors.

6.2 Ordering data. When acquisition documents are invoked, they should specify the following:

a. Title, number, and date of this handbook.

b. Technique procedure (see 1.2).

c. Issue of ASSIST to be cited in the solicitation, and, if required, the specific issues of individual documents referenced (see 2.2, 2.3, and A.2.2).

d. Foam ingredients approved for the specified procedure (see 4.1.1).

e. Electrostatic discharge sensitive items permitted (see 4.2).

f. Heat sensitive items permitted (see 4.2).

g. Method of preservation required (see 4.4).

h. Minimum foam thickness requirement (see 4.7).

i. Rough handling test requirement (see 4.7 and appendix A, A.3.5.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

j. Complete container requirements, including specifications, classification, and dimension (see 5.1).

k. Class and grade/category of foam material (see 5.1.1, 5.1.2, 5.1.3, 5.1.4, and 5.1.8).

l. Width (diameter) of plastic film bag (or tube), technique IV (see 5.1.4.1).

m. Minimum thickness (mil) of plastic sheet (see 5.1.4.1).

n. For technique V, include: (1) container specification and size; (2) foam material in accordance with MIL-PRF-83671; (3) re-required design steps; (4) associated wraps or other materials; (5) all figures, illustrations, and drawings necessary to fully describe the special design techniques; (6) special markings (see 5.1.5).

o. Complete description to include chemical class and grade, associated materials, and applicable drawings (see 5.1.6 and 5.1.7).

p. Design procedures, technique VII (see 5.1.7.1).

q. When containers are fabricated from material other than specified (see 5.1.7.2).

r. Minimum clearance requirements, technique VII (see 5.1.7.2).

s. Amount of foam requirements, technique VII (see 5.1.7.2).

t. Established fragility level for shock-sensitive items (see 5.2.3).

u. Pack conditioning requirement (see 5.2.9).

v. Additional special marking, as required (see 5.4.1)

w. Group B sampling and acceptance procedures and Group C quality conformance tests and examinations and all inspection requirements (see A.3.4.1, A.3.4.1.1, A.3.4.1.3, A.3.4.1.4, and A.3.4.1.4.1).

x. Container tests, as required (see A.3.5.3.2)

6.3 Sampling for Group B inspection. As a guide for Group B inspection, the sampling plan in MIL-STD-1916, DoD Preferred Methods for Acceptance of Product or ANSI/ASQ Z1.4, Sampling Procedures and Tables for Inspection by Attributes may be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

6.4 Subject term (key word) listing.

Encapsulated pack
Foam-in-bag
Flexible foam
FIP packaging
Inverted pack
Preformed molds
Rigid foam
Semirigid foams
Split packs
Urethane foam

6.5 Identification of changes. Marginal notations are not used in this handbook to identify changes from the original specification document due to the extensiveness of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

1. Determine the static-bearing stress (pounds per square inch (psi) for each surface (weight/each surface area that will be bearing on the cushion).

2. Next, determine the item's fragility measured in g's (obtained from manufacturer or engineering estimate).

3. The thickness of foam needed may be found by looking at the cushion curves. All curves are for 30-inch drop.

4. To find foam thickness for each face of item, draw a line upward from horizontal of axis at point of static stress for that item face (1).

5. Follow a line drawn from the left (2) to a point on the vertical axis equal to the fragility estimated earlier.

6. The point (3) when these two lines cross will be near one of the cushion curves on the graph.

7. The closest cushion curve below crossing point will be the thickness of foam needed to protect face of the item in a 30-inch drop.

8. Repeat process for each of the other faces of the item. Use these values for each face or use the thickest for all faces for extra protection.

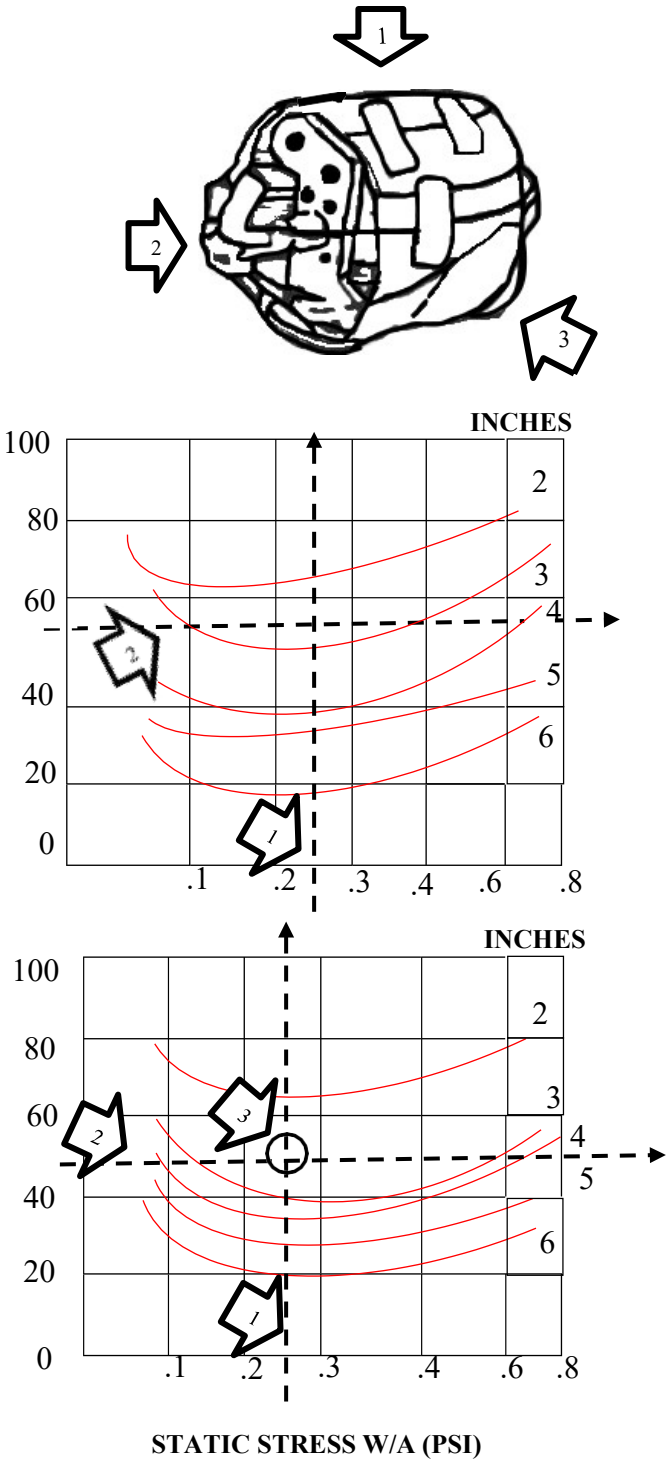


FIGURE 1. **DESIGN FOR FLEXIBLE FOAM PACKS.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

9. Do not package item in flexible foam if the point where the two lines cross is below the curves.

10. The optimum loading point is indicated by the arrow. This point is defined as the lowest shock level that can be achieved using a particular 2-inch thickness cushion. Obviously then, the optimum static stress is defined by the vertical line. The problem really begins when a calculated static stress is greater than the optimum value. This could be the case in the example shown on the curve (indicated by A). In this case, item fragility was determined to be 60 g's and the calculated static stress is 0.5 if the "closest cushion curve below the crossing point" is assumed to be the 2 inch, there is a dangerous condition existing. The cushion is overloaded; such a situation allows serious loss of thickness of the cushion and increasing shock loads during subsequent drops. Two options are available; i.e., (1) change static stress to a lower value by adding support pieces to spread the load, or (2) go to the next lower cushion curve wherein the crossing point occurs either at or to the left of the optimum point.

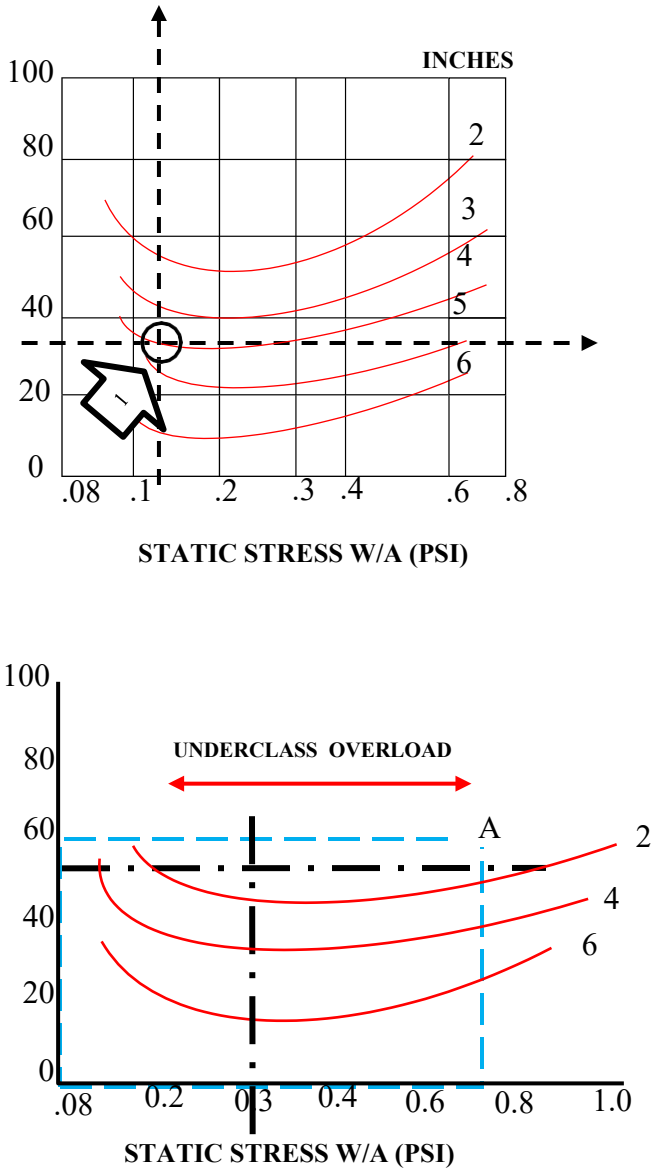


FIGURE 1. Design for flexible foam packs – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

2 in/5.1 cm		3 in/7.6 cm		4 in/10.2 cm	
Maximum Weight (lb./kg)	Minimum allowable bearing surface (sq. in/sq. cm)	Maximum Weight (lb./kg)	Minimum allowable bearing surface (sq. in/sq. cm)	Maximum Weight (lb./kg)	Minimum allowable bearing surface (sq. in/sq. cm)
1/0.5	2.5/15	26/11.8	51/340	51/23.1	85/550
2/0.9	5.0/15	27/12.2	54/350	52/23	85/550
3/1.4	7.5/30	28/12.7	56/360	53/24.0	90/580
4/1.8	10.0/65	29/13.2	58/370	54/24.5	90/580
5/2.3	12.5/80	30/13.6	60/390	55/25.0	90/580
6/2.7	15.0/95	31/14.0	62/400	56/25.4	95/610
7/3.2	17.5/115	32/14.5	64/410	57/25.9	95/610
8/3.6	20.1/130	33/15.0	66/430	58/26.3	95/610
9/4.1	22.5/145	34/15.4	68/440	59/26.8	100/645
10/4.5	25.0/160	35/15.9	70/450	60/27.2	100/645
11/5.0	27.5/175	36/16.3	72/460	61/27.7	100/645
12/5.4	30.0/195	37/16.8	74/480	62/28.1	105/680
13/5.9	32.5/210	38/17.2	76/490	63/28.6	105/680
14/6.4	35.0/225	39/17.7	78/500	64/29.0	105/680
15/6.8	37.5/240	40/18.1	80/520	65/29.5	110/710
16/7.3	40.0/260	41/18.6	82/530	66/29.9	110/710
17/7.7	42.5/275	42/19.1	84/540	67/30.4	110/710
18/8.2	45.0/290	43/19.5	86/550	68/30.8	115/740
19/8.6	47.5/305	44/20.0	88/570	69/31.3	115/740
20/9.1	50.0/320	45/20.4	90/580	70/31.8	115/740
21/9.5	52.5/240	46/20.8	92/590	71/32.2	120/770
22/10.0	55.0/355	47/21.3	94/610	72/32.7	120/770
23/10.4	57.5/370	48/21.8	96/620	73/33.1	120/770
24/10.9	60.0/385	49/22.2	98/630	74/33.6	125/810
25/11.3	62.5/405	50/22.7	100/645	75/34.0	125/810

FIGURE 2. Design of 0.5 pound density semirigid foam packs.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

5 in/12.7 cm		6 in/15.2 cm	
Maximum Weight (lb./kg)	Minimum allowable bearing surface (sq. in/sq. cm)	Maximum Weight (lb./kg)	Minimum allowable bearing surface (sq. in/sq. cm)
76/34.5	110/710	104/47.2	140/900
77/34.9	110/710	108/49.0	140/900
78/35.4	110/710	112/50.8	140/900
79/35.8	115/740	116/52.7	150/970
80/36.3	115/740	120/54.4	150/970
81/36.7	115/740	124/56.2	160/1040
82/37.2	115/740	128/58.0	160/1040
83/37.6	120/770	132/59.8	170/1100
84/38.1	120/770	136/66.7	170/1100
85/38.6	120/770	140/63.5	180/1170
86/39.0	125/810	144/65.3	180/1170
87/39.5	125/810	148/67.1	190/1230
88/40.0	125/810	152/68.9	190/1230
89/40.4	125/810	156/70.8	200/1290
90/40.8	130/840	160/72.6	200/1290
91/41.3	130/840	164/74.4	210/1360
92/41.7	130/840	168/76.2	210/1360
93/42.2	135/870	172/78.0	220/1420
94/42.6	135/870	176/79.8	220/1420
95/43.1	135/870	180/81.6	230/1490
96/43.6	135/870	184/83.5	230/1490
97/44.0	140/900	186/84.4	240/1550
98/44.4	140/900	190/86.2	240/1550
99/44.9	140/900	194/88.0	240/1550
100/45.4	140/900	200/90.7	260/1680

NOTE: If these parameters cannot be met, this foam material composition should not be used.

1/ Item weight includes total weight ready for foaming, i.e., item plus wraps, barriers, tapes, pads, adhesives, straps, etc. When the weight falls exactly between two successive weight ranges, use the higher figure.

2/ When minimum allowable bearing surface area falls exactly between two successive figures, use next highest figure.

NOTE: When using semirigid (semiflexible) foams, bearing surfaces can be improved (increased), as necessary, by affixing them to platform, containerizing, etc., then foaming, as required.

FIGURE 2. Design of 0.5 pound density semirigid foam packs – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

1. Determine the tube/bag width by applying the following formula:

FOR FLEXIBLE FOAM

$$\text{Tube/bag width (inches)} = \frac{M/N}{IW \times 4} + FT$$

FOR RIGID FOAM

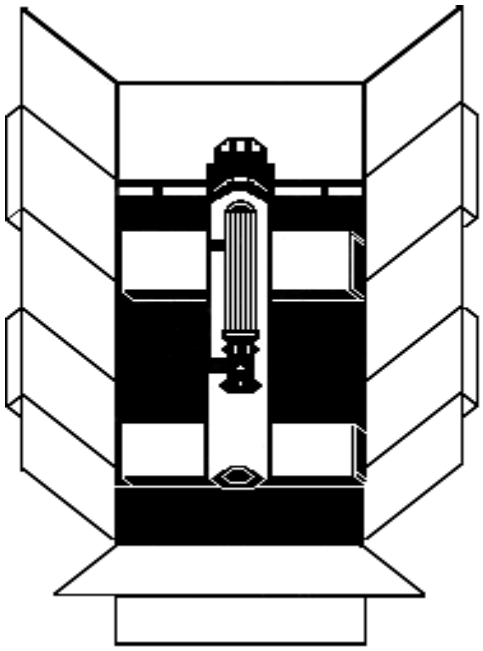
$$\text{Tube/bag width (inches)} = \frac{M/N}{IW \times 20} + FT$$

M = Weight of prepared item (lb.)
N = Number of bags to be used
IW = Width of item (inches)
FT = Desired/required foam thickness (inches)

2. Round off the calculated tube/bag width to the nearest inch.

3. If tube/bag width is calculated to be less than 12 inches, use a 12-inch tube/bag.

4. If the calculated tube/bag width exceeds 18 inches, consider increasing the number of tubes/bags.



FOR RIGID FOAMS ONLY:

Bag width should at no time be less than one and one-half times the desired/required foam thickness.

If calculated tube/bag width is less than one and one-half times the foam thickness, select a wider tube/bag to meet the one and one-half minimum.

FIGURE 3. Determination of tube/bag width for technique IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

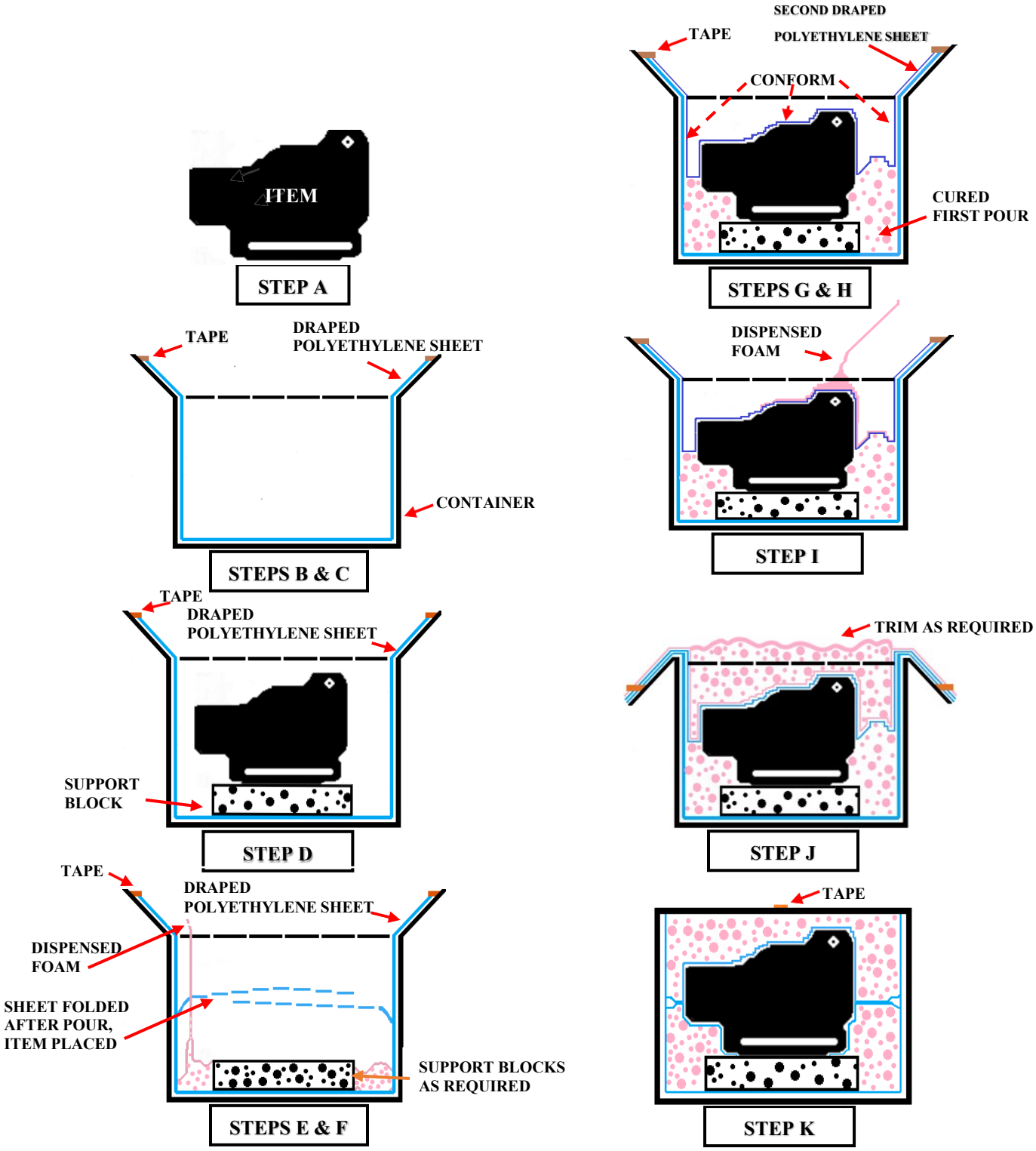


FIGURE 4. Split pack, standard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

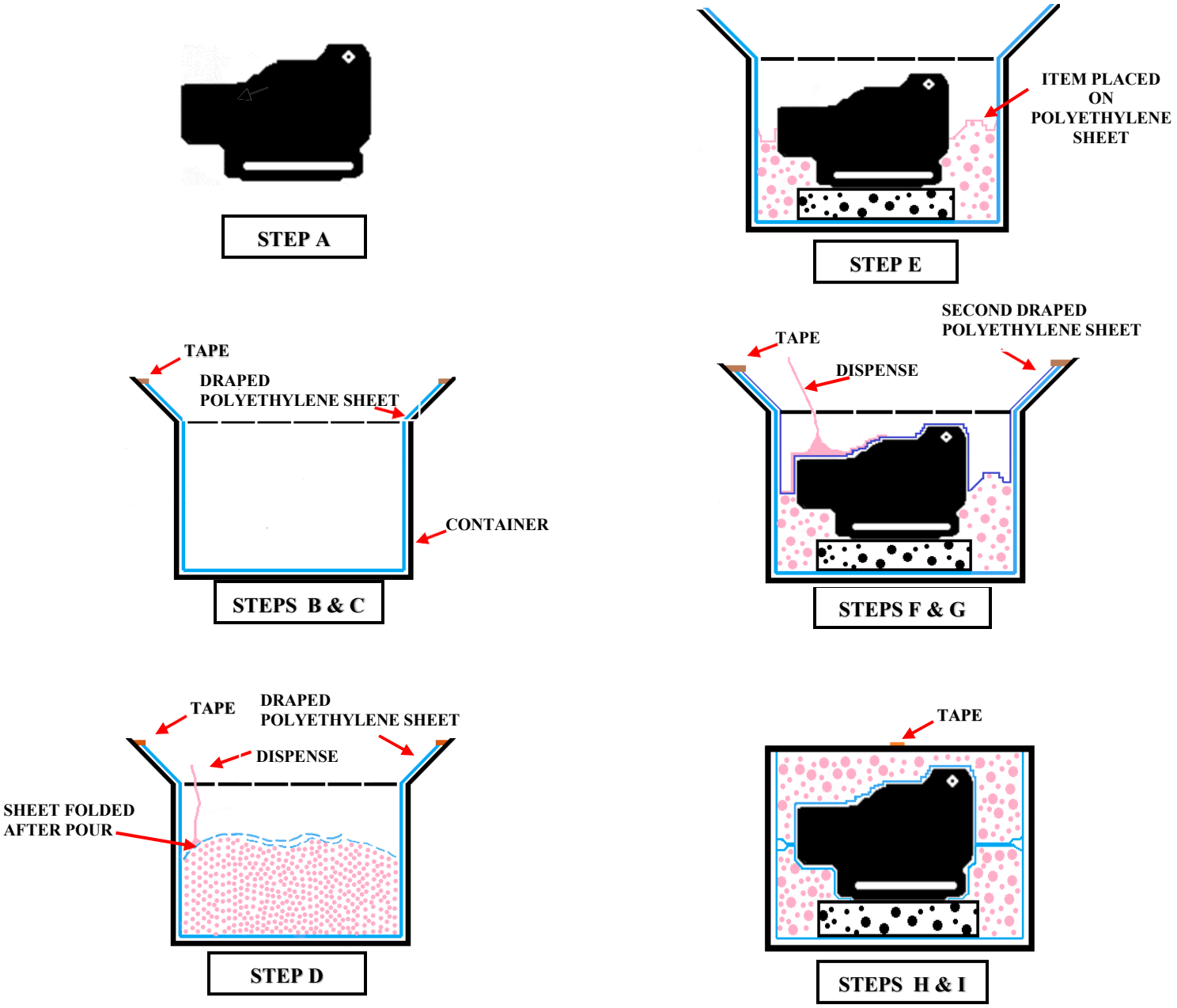


FIGURE 5. Split pack, alternate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

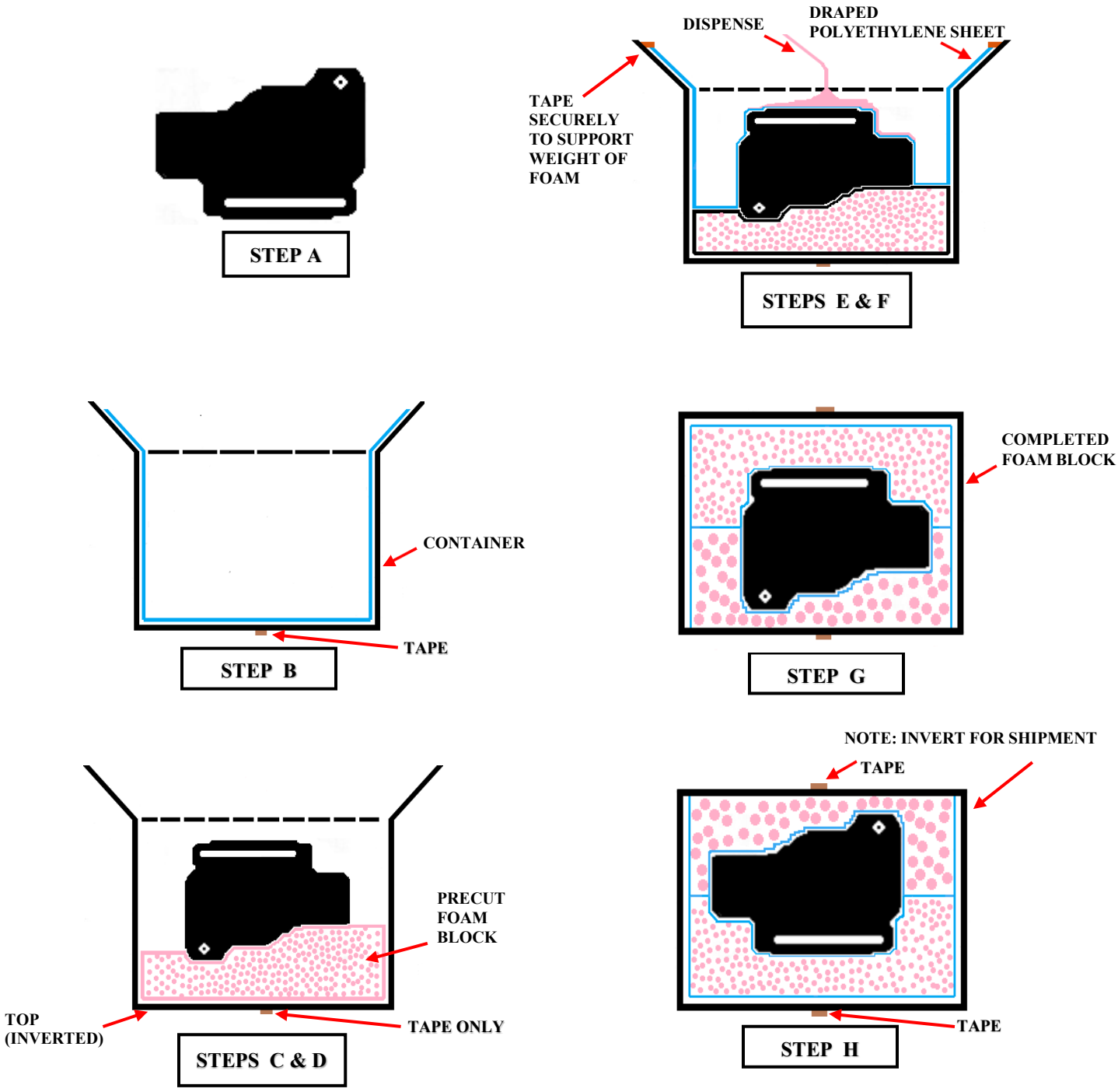


FIGURE 6. Split pack, inverted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

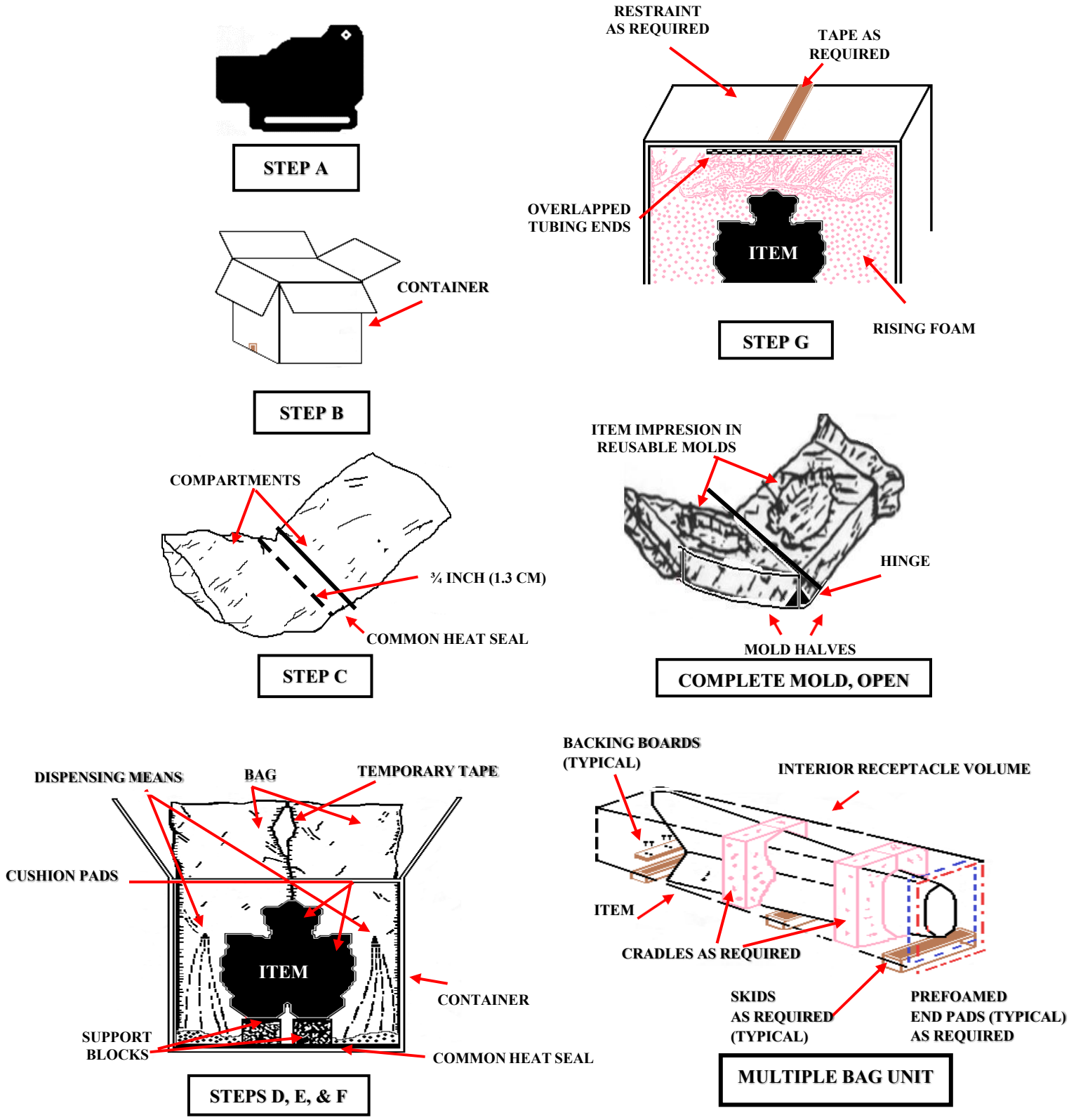


FIGURE 7. Foam-In-Bag.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

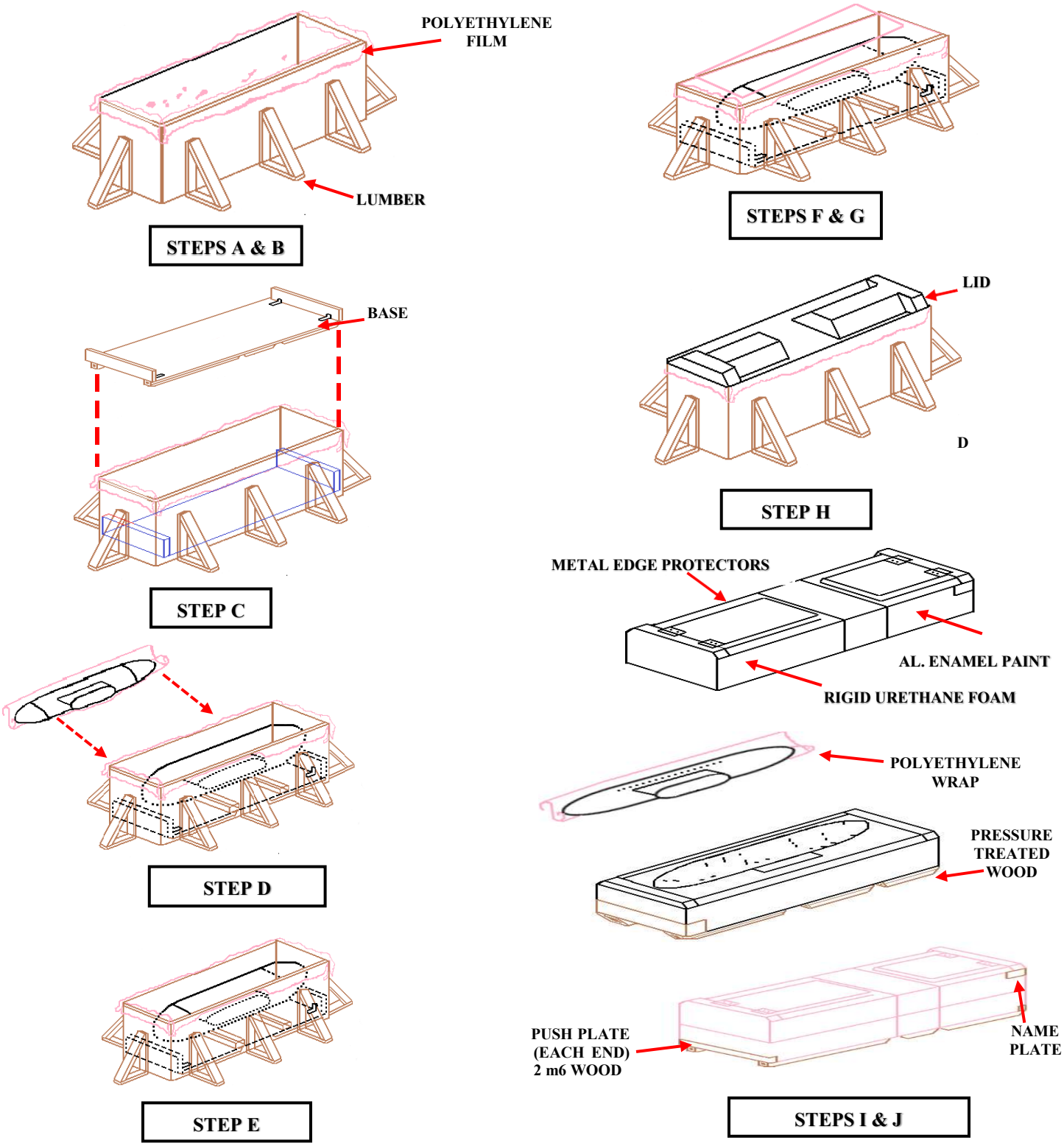


FIGURE 8. Foam container.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

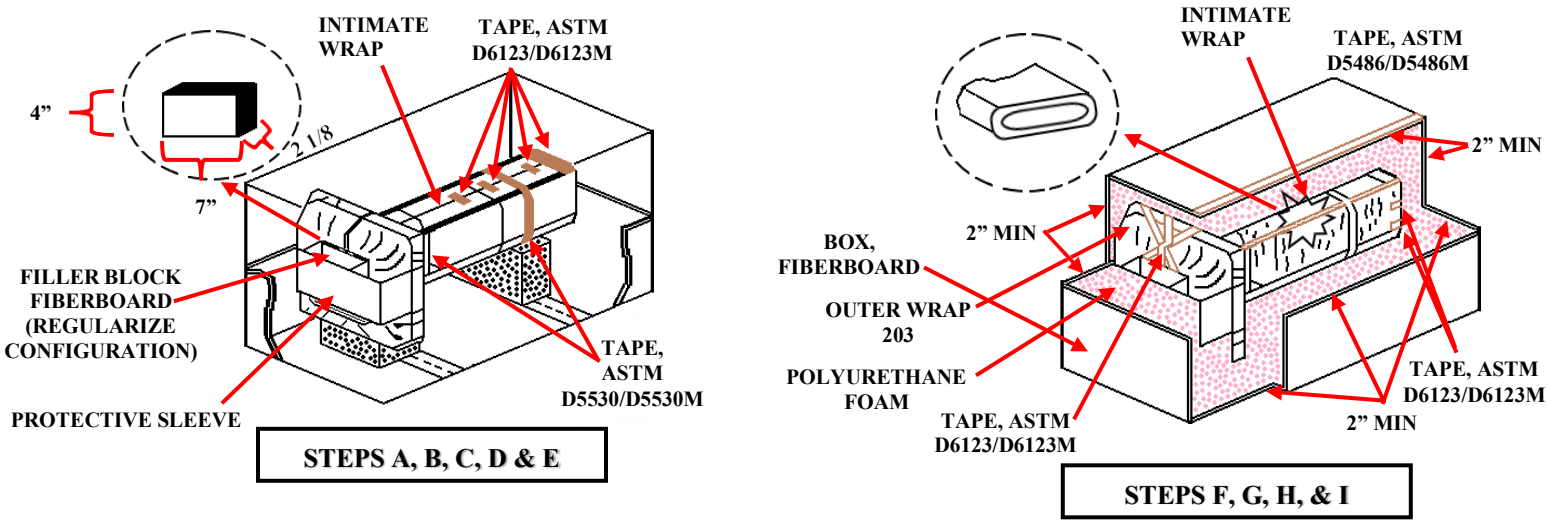


FIGURE 9. Encapsulated pack (Design 1).

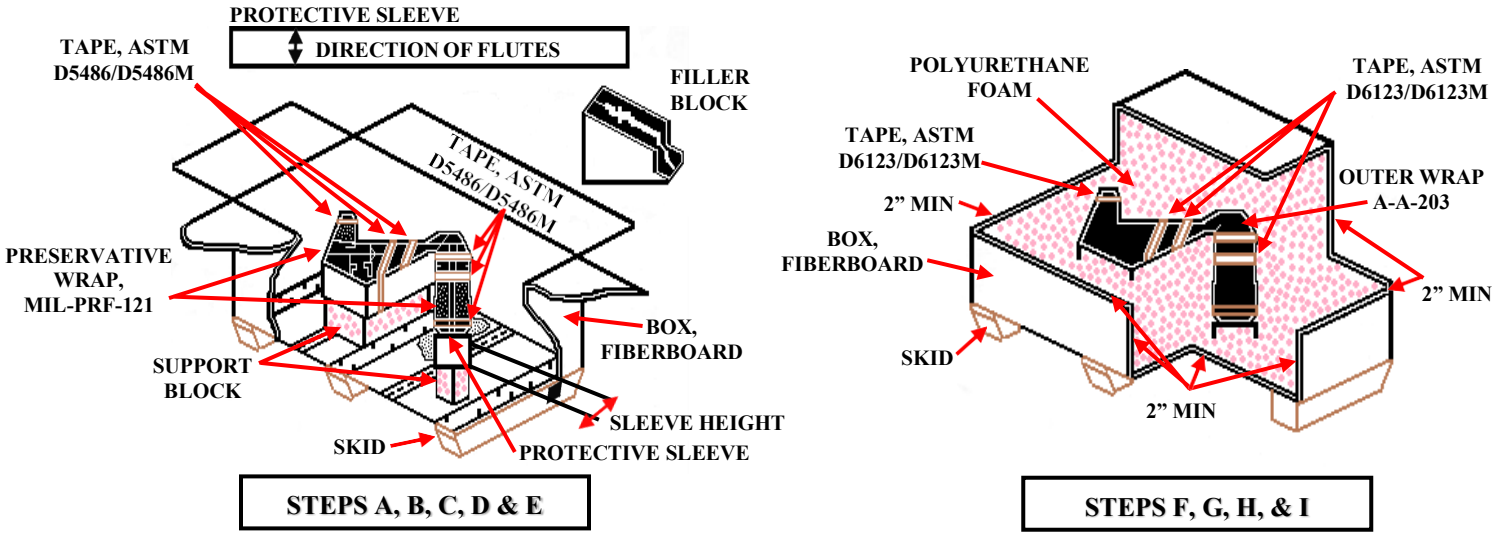


FIGURE 10. Encapsulated pack (Design 2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

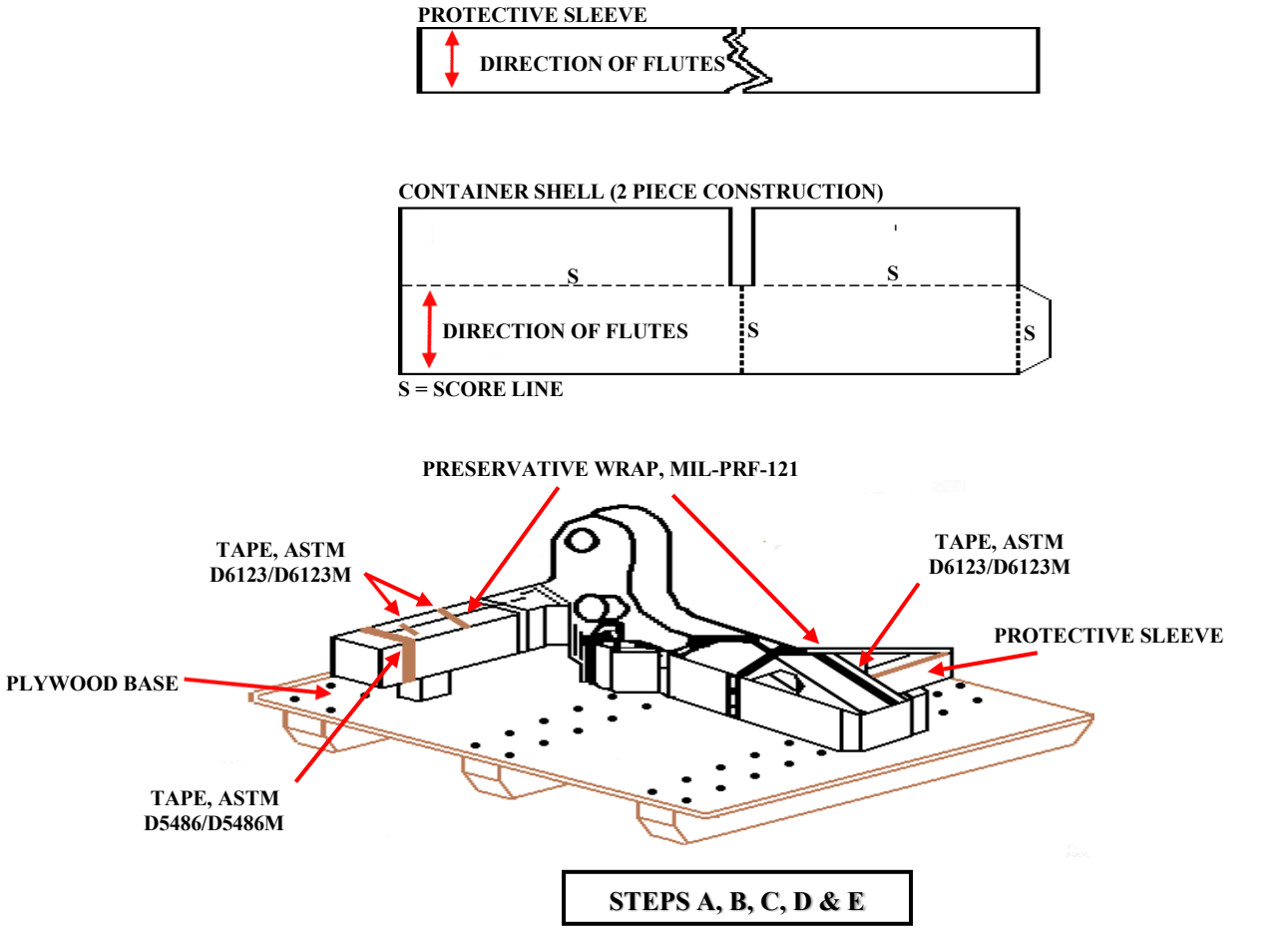


FIGURE 11. Encapsulated pack (Design 4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

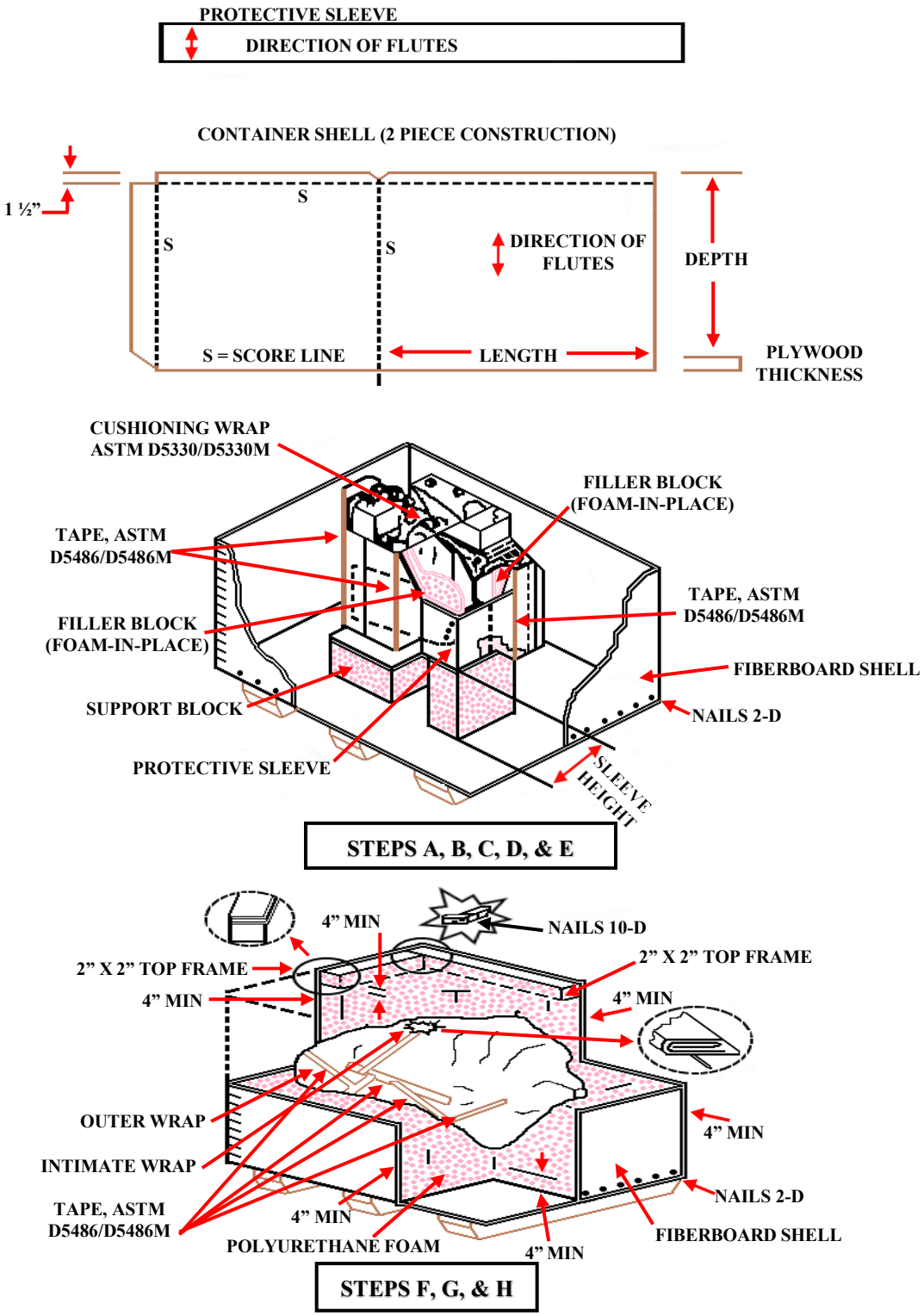


FIGURE 12. Encapsulated pack (Design 4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

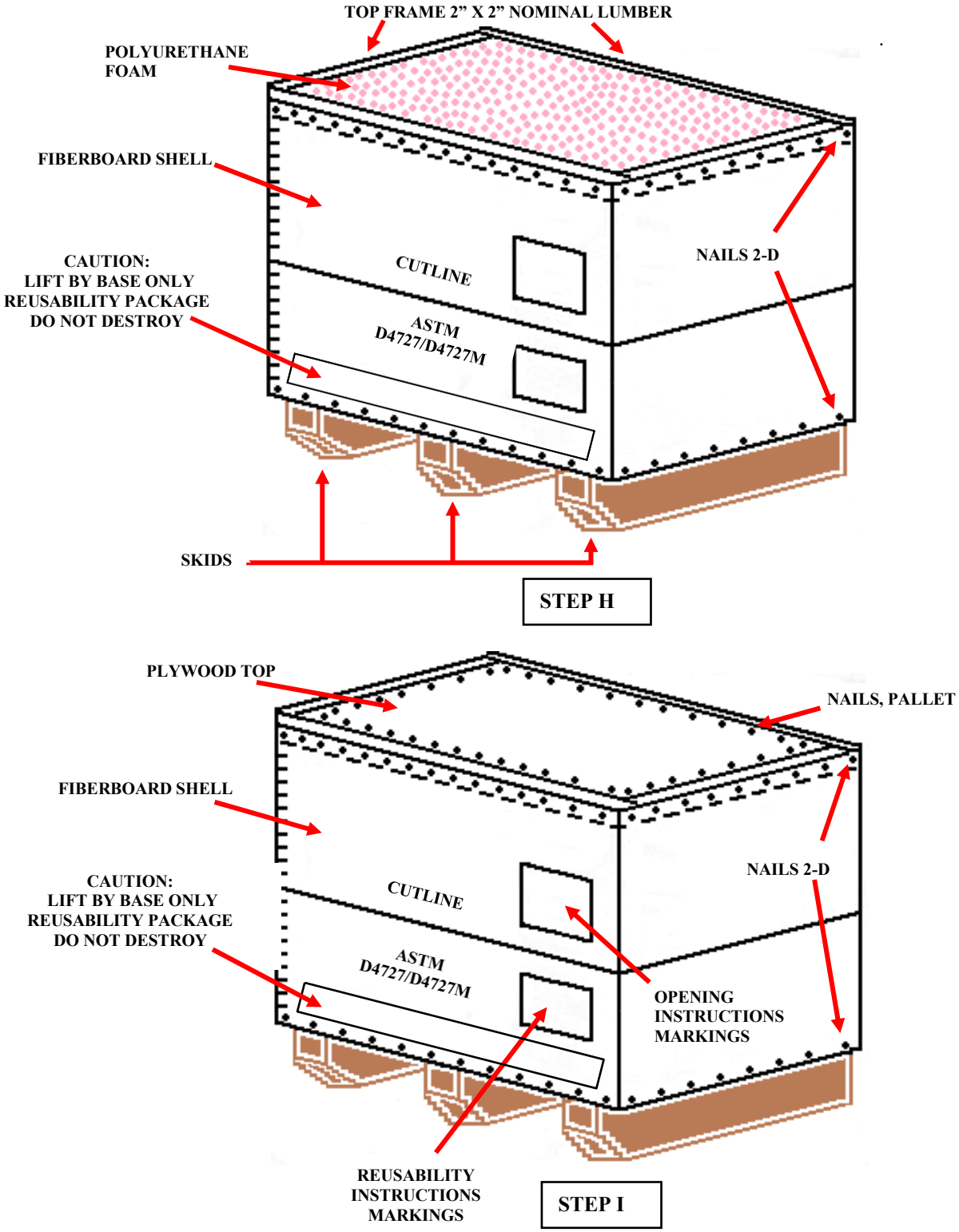


FIGURE 12. Encapsulated pack, (Design 4) – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

OPENING INSTUCTIONS:

1. CONTAINER SHOULD BE OPENED BY CUTTING ALONG "CUT LINE" USING EITHER OF THE FOLLOWING METHODS:

A. MAKE INITIAL CUT WITH PORTABLE POWER SAW WITH BLADE SET AT 1-INCH DEPTH. CUT THROUGH THE REMAINDER OF THE POLYURETHANE FOAM WITH A HAND SAW OR LONG BLADE KNIFE.

B. USE A HAND SAW, PREFERABLY RIP-TYPE, MEDIUM BLADE.

2. USING EITHER METHOD, THE CUT SHOULD BE CONTINUOUS AROUND THE PERIMETER OF THE CONTAINER, WHEN USING A HAND SAW, CUT THROUGH THE CONTAINER SHELL AND INTO THE FOAM USING A SABRE SAW CUTTING TECHNIQUE UNTIL THE BLADE REACHES THE PROTECTIVE SLEEVE. (CAUTION: DO NOT JAM BLADE THROUGH SLEEVE AS ITEM MAY RECIEVE DAMAGE.)

3. AFTER CUTTING, REMOVE TOP HALF OF CONTAINER, TO FACILITATE REMOVAL, TWO FLAT STEEL STRIPS OR ANGLE IRONS 3 INCHES WIDE AND OF SUFFICIENT LENGTH TO EXTEND A MINIMUM OF 12 INCHES BEYOND THE ENDS OF THE CONTAINER SHOULD BE INSERTED INTO THE CUT ALONG THE ENDS TO EFFECT THE INITIAL SEPARATION. EXERT PRESSURE AT BOTH ENDS SIMULTANEOUSLY.

4. REMOVE THE ITEM FROM THE LOWER HALF OF THE CONTAINER EXERCISING CAUTION NOT TO FRACTURE THE FOAM.

REUSABILITY INSTRUCTIONS:

ONCE OPENED, CONTAINER SHOULD BE REUSED FOR SHIPMENT OF THE UNSERVICEABLE ASSET TO REPAIR OR REBUILD ACTIVITY.

1. PLACE ITEM IN CAVITY OF BOTTOM SECTION OF CONTAINER.

2. POSITION TOP SECTION OF CONTAINER OVER ITEM. (CAUTION: FILL VOID AREAS WITH CELLULOSE WADDING. IF LARGE VOIDS EXIST DUE TO FOAM REMOVAL, DO NOT REUSE.)

3. SECURE TOP TO BOTTOM BY PLACING STEEL STRAPS CONFORMING TO ASTM D3953 AROUND SIDES, TOP, AND BOTTOM OF CONTAINER. TIGHTEN STRAPS AND SECURE WITH STRAP SEALS.

4. REMOVE OR OBLITERATE OBSOLETE MARKINGS AND APPLY APPROPRIATE MARKINGS.

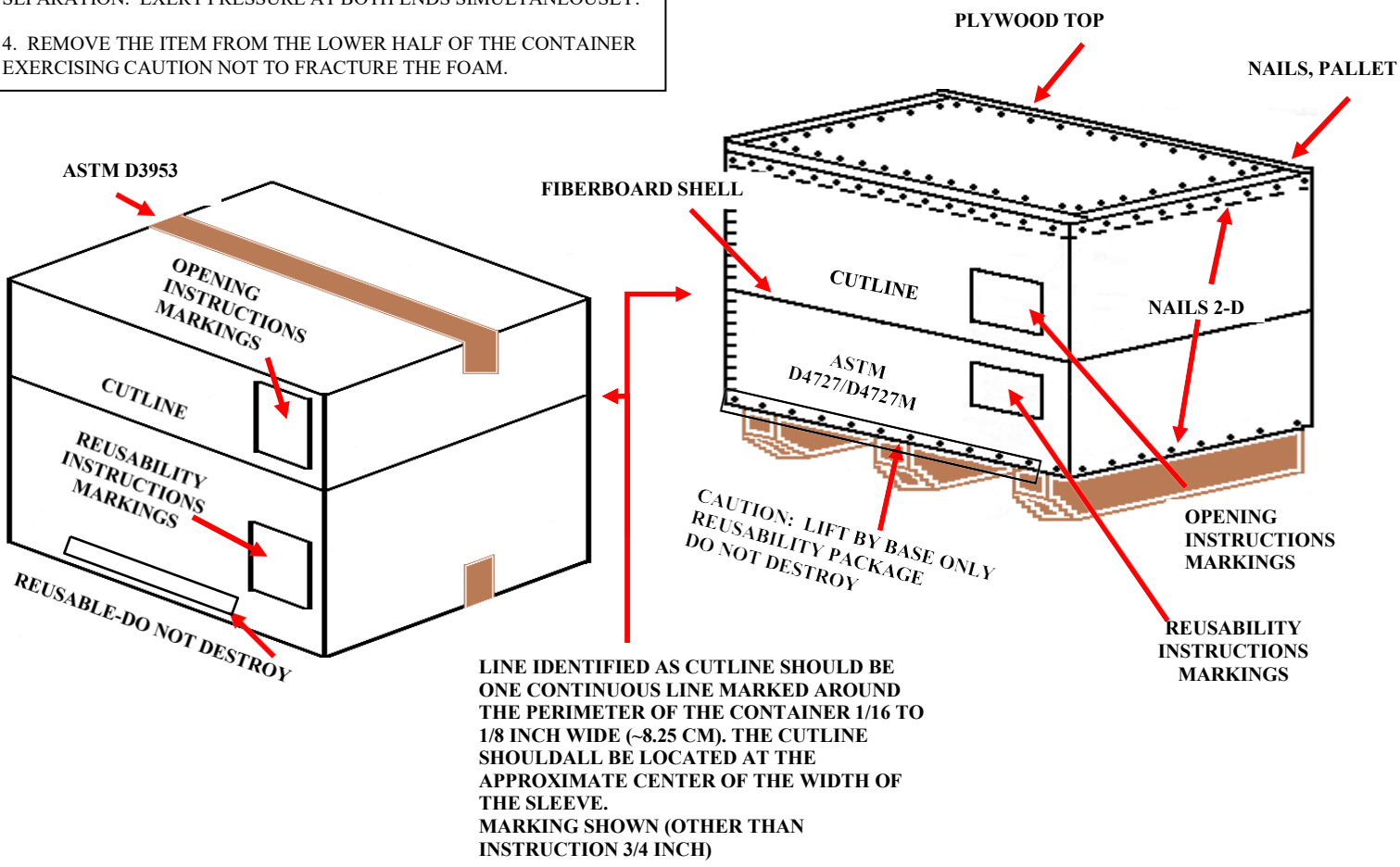
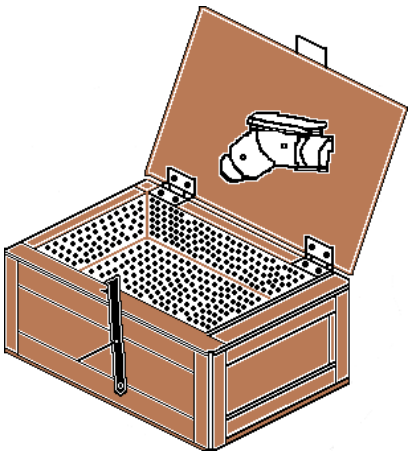
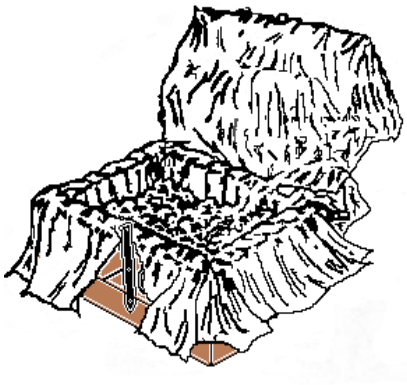


FIGURE 13. Encapsulated pack, special marking locations.

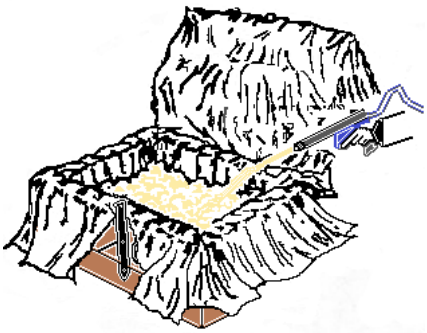
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.



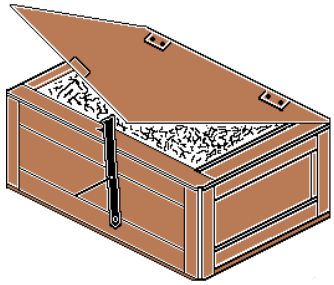
STEP A



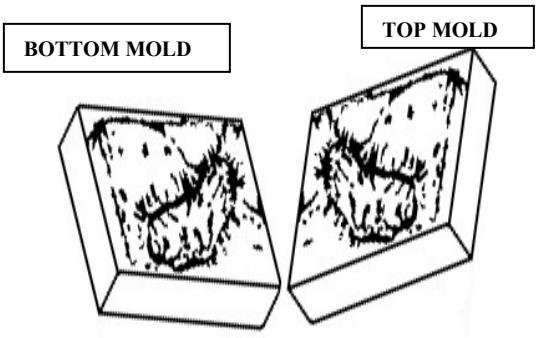
STEP B



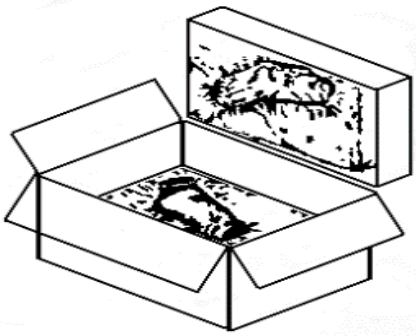
STEP C



STEP D



STEPS E & F



STEP G

FIGURE 14. Preformed Mold.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

APPENDIX A

QUALITY ASSURANCE PROVISIONS

A.1 SCOPE.

A.1.1 Scope. This appendix includes and classifies all inspections, examinations, and tests necessary for determining that the requirements and procedures specified in sections 4 and 5 are being adhered to properly. It provides procedural and sampling guidance for accomplishment of that compliance, assigns responsibilities for the same, and specifies quality conformance standards.

A.2 APPLICABLE DOCUMENTS.

A.2.1 Government documents.

A.2.2 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this appendix to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Acquisition Streamlining and Standardization Information System (ASSIST) and supplement thereto, cited in the solicitation (see 6.2).

MILITARY

MIL-STD-3010 - Test Procedures for Packaging Materials.

A.3 QUALITY ASSURANCE PROVISIONS.

A.3.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this handbook where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

A.3.1.1 Responsibility for compliance. All items should meet all requirements of sections 4 and 5. The inspections set forth in this handbook should become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the handbook should not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

Government for acceptance comply with all requirements of the contracts. Sampling inspection, as part of manufacturing or service operation, is an acceptable practice to ascertain conformance to requirements. However, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

A.3.1.2 Alternate test procedure approval. In instances where a test may necessitate an impractical manipulation of a mounted item, where the overall size or weight of the item or test equipment availability obviates compliance with a specific test requirement of this specification, the contractor may submit a request accompanied by detailed justification through the cognizant Government inspector for approval of an alternate test procedure.

A.3.2 Classification of inspections. The inspections specified herein are classified as quality conformance inspections. (see A.3.4)

A.3.3 Inspection conditions. All inspections should be performed under ambient environmental conditions of temperature and relative humidity. Best results are obtained when tested in a temperature range between 68° F. (20° C.) and 85° F. (29° C.) with a relative humidity of 40 to 60 percent. Regardless of the procedures being used, no foam packs should be either examined or tested prior to 24 hours after final pouring of the polyurethane ingredients.

A.3.4 Quality conformance inspection.

A.3.4.1 Inspection of product for delivery. Inspection of product for delivery should consist of group A, B, and C inspections. Sequence of inspections should be exactly as specified herein. This will ensure that procedural steps have been followed correctly and end-product foam characteristics achieved without adverse effects on items, other protective packaging materials, or containers (see 6.2).

A.3.4.1.1 Inspection lot. An inspection lot should consist of all foam packs of the same technique produced under essentially the same conditions and offered for inspection at one time (see 6.2).

A.3.4.1.2 Group A inspection. Group A inspection should consist of certification supported by verifying data specified in table I in the order shown. Sampling and acceptance procedures should be as specified in MIL-STD-2073-1.

A.3.4.1.3 Group B inspection. Group B inspection should consist of the examinations and test specified in table I in the order shown. Selection and use of sampling and acceptance procedures should be specified in the contract or order (see 6.2). Guidance for those procedure for group B inspection is given in 6.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

A.3.4.1.3.1 Disposition of sample units. Sample units that have been subjected to group B inspection should not be delivered on the contract or purchase order.

A.3.4.1.4 Group C inspection. Group C inspection should consist of the examinations and tests specified in table I, as required, for the procedure and foam formulation used.

A.3.4.1.4.1 Sampling plan. When group C tests are specified in the contract or order for quality conformance inspection, samples should be selected at random and subjected to group C tests. Sampling and acceptance procedures should be specified in the contract or order (see 6.2).

A.3.4.1.5 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects or screen out the defective units and resubmit for reinspection. Resubmitted lots should be inspected using tightened inspection in accordance with MIL-STD-1916 or ANSI/ASQ Z1.4. Such lots should be separate from new lots and should be clearly identified as reinspected lots. With technique VII, the entire lot is not rejected if a single failure occurs. Instead, the next five packs produced after the failure should be examined. If failure repeats in any of these packs, the entire lot should be rejected. Also, for technique VII, only every fourth lot should be examined. Acceptance of the inspected lots presumes acceptance of the lots between them. For example, if lots 4 and 7 are accepted, then lots 5 and 6 are also accepted.

A.3.5 Methods of inspection.

A.3.5.1 Examinations of packs. Group A examinations should be affirmed by certification from the contractor. Group B examinations should be visual inspections or tests, as required. Failure to comply with all of the visual examinations and test certification requirements should be cause for rejection (table I).

A.3.5.2 Inspection for surface distortion. Packages should be inspected for surface distortion characterized by split closure tape, bulging container walls, or container concavity. The degree of surface distortion should be determined using a straight edge of appropriate length for the particular dimensions of the pack. All surfaces, excluding bottom surfaces where skids are attached, should be checked for a measurable distortion, as determined by placing the straight edge along the diagonal of each surface (see 5.2.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

TABLE I. Tests Required for the Procedure and Foam Formulation Utilized.

GROUP A	Require para	
-Designated packaging materials and FIP ingredients used in the fabrication of the foam pack.	4.1 thru 4.1.3	
-Required foaming equipment and supplies selected for development of foam pack.	4.5	
-Preparation for developing foam packs – acceptable.	4.6 thru 4.8	
-Foaming technique followed for the applicable procedure.	5.1	
-Opening method suitably described on pack exterior.	5.4.1	
-Environmental considerations pertinent to conditions affecting operating personnel, ecology, and waste management – acceptable.	4.3	
GROUP B	**Require para**	**Test para**
- Reliability of opening method.	5.2.4	A.3.5.5
- Completeness of void fill, as required.	5.2.5	A.3.5.5
- No evidence of foam breakthrough and undesirable adherence.	5.2.6	A.3.5.5
- Complete release of item from foam pack.	5.2.7	A.3.5.5
- No evidence of item damage, foam particles, static charges, or unsafe humidity indicators.	5.2.8	A.3.5.5
- Reusability.	5.3	A.3.5.8
- Marking and labeling of foam packs properly applied.	5.4	A.3.5.6
- Foam workmanship.	5.5	A.3.5.7
GROUP C	**Require para**	**Test para**
- Surface distortion.	5.2.1	A.3.5.2
- Rough handling test.	5.2.2	A.3.5.3
- Shock mitigation (flexible foams).	5.2.3	A.3.5.4

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

A.3.5.3 Rough handling test. Pack should be subjected to the following sequence of rough handling tests in accordance with the applicable test method from MIL-STD-3010. Test Method 5011 addresses the rough handling test.

A.3.5.3.1 Small containers test. Those having no edge or diameter over 60 inches (152.40cm) and a gross weight of 150 pounds (68.10 kg) or less. Any container with a gross over 150 pounds or a length of any edge over 60 inches should be tested as a large container. Additionally, any container equipped with skids should be tested as a large container.

Test Method 5007 Free Fall Drop Test, Procedure B
Test Method 5007 Free Fall Drop Test, Procedure E

A.3.5.3.2 Tests for large containers and all others.

Test Method 5005 Cornerwise Drop (Rotational) Test.
Test Method 5008 Edgewise Drop (Rotational) Test
Test Method 5019 Vibration (Repetitive Shock) Test (when specified, see 6.2)
Test Method 5023 Incline-Impact Test (when specified, see 6.2)

Graduated drop and impact test heights and number of impacts should be in accordance with table IV of MIL-STD-2073-1, as determined by the gross weight of the container and contents. Following the performance of these tests, exterior surfaces of the pack should be inspected for evidence of container shell, base, and top frame separation from the polyurethane foam for those procedures not using intentional parting-type materials. Evidence of this separation should be cause for rejection of the pack. Determination of conformance to the item looseness will be made following opening. Evidence of the item movement in any direction; crushing of foam materials: item rotation from original placement; or item damage such as broken or distorted projections, loosened mounting, broken wires, or scorched surfaces should be cause for rejection.

A.3.5.4 Examination for shock mitigation (flexible foams). Packs should be opened and examined initially for conformance to 5.2.4 through 5.2.9 and the test specified in A.3.5.5. Prior to determining reusability, the item should be removed, as necessary, to permit attachment of appropriate accelerometers. Three accelerometers, one for each major axis or a single triaxial unit, should then be mounted on the item or simulated item as close to the center of gravity as possible, in such a manner to ensure against improper movement during subsequent shock tests. Cable leads should then be attached to the sensors and the pack reclosed. During the reclosure, the pack should be examined for conformance with the requirement in 5.5 and the test specified in A.3.5.7.

Access through the container wall should be accomplished for extending the sensor cable leads to the instrumentation. Readout provisions should consist of necessary measuring instruments to include any calibration reference, amplifiers, or recorders that will ensure

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

an accuracy of +5 percent in the readings obtained. Free fall, flat drop tests should be performed initially at ambient temperatures in accordance with Procedure B, Test Method 5007, of MIL-STD-3010, using the graduated drop heights of table I of that standard for level A requirements. A series of six drops should be made on each flat surface of the container with only the final five drops being recorded. From the reading taken for each drop, the resultant "g's" should be determined for that drop. The average of the five resultant shocks should be compared with the required "g" protection. A higher "g" figure than that allowed should be cause for rejection of the pack. The identical pack should be conditioned for 24 hours at a temperature of -20° F. (-29°C) (see 5.2.9). Immediately following conditioning of the pack, two additional drops should not exceed the specified fragility level. Failure should be cause for rejection of the pack (see 5.2.3).

A.3.5.5 Opening test. All techniques of FIP packs should be opened in accordance with instructions provided on the pack exterior (see 5.4.1) or for those packs in which opening is obvious. For purposes of this test, opening should include those operations required to provide access to the prepared item and those visual examinations necessary to confirm the acceptability of the procedure (see 5.2.5 through 5.2.8). The interior of the pack should then be examined for degree of void fill, improper foam adherence or loose particles (breakthrough), difficult item removal (locking or sealing), humidity indicator in a safe condition, and suspected damage to items. A minimum of 95 percent fill of the intended void space should exist. Where the specified procedure permits, the foam exterior surfaces should be examined for void areas (see 5.2.5).

A.3.5.6. Marking and labeling examination. Visual examination of the exterior of the completed package should verify the proper application of any marking and labeling, as specified (see 5.4).

A.3.5.7 Foam workmanship examination. The FIP packs should be examined for evidence of splits, sparklers, shrinkage, voids, irregularities, or striations in the foam (caused by incomplete foam mix). Sample blocks should be cut from the foam packs(s) and trimmed to provide plane and parallel surfaces to a size of 4 inches (10.16 cm) by 4 inches (10.16 cm) by 2 inches (5.08 cm). The thickness dimension should be perpendicular to the direction of rise. One specimen should be taken from each of two opposite sides. For purpose of this test, halves formed constitutes a single block. When two or more successive pours (as opposed to a single, continuous one) are made, thereby creating a fusion line where the pours meet, then the sample specimen should be cut so that the fusion line is included across the specimen as near to its center or midpoint as possible. This will allow for examination of the quality of the "knit" of these pours. The examined specimens should be free of soft or tacky particles (unexpended resin) or show evidence of splits, sparklers, or shrinkage. Further, there should be no void openings or pockets (blow holes) exceeding 0.5 inch in any direction or evidence of scorching during foam formation. Packs utilizing reused foam should be examined for incomplete bonding of refoamed materials caused by using excessive amounts of previously foamed materials

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

without enough virgin foam around these materials to provide a complete bond (see 5.5). If examination reveals any of the aforementioned characteristics, it should be cause for rejection.

A.3.5.8 Reusability. Formed foam mold packs should be tested to ensure that, once opened as in A.3.5.5, the molds formed allow ready replacement of the prepared item and reuse of the pack. Failure to accomplish these requirements should be cause for rejection (see 5.3). This applies to packs other than technique VII packs. Technique VII opened for quality conformance inspection should not be reused or repaired except for return of an unserviceable part (see 5.1.7.4, 5.2.4, and 5.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

APPENDIX B

ENGLISH-METRIC CONVERSIONS

B.1 SCOPE. This appendix gives the necessary metric conversion for each English equivalent used throughout this handbook. This is in addition to equivalents already given parenthetically within the proceeding section. Standard measurement equivalents and supplemental equivalents are given here also to assist in converting any computations that result from the various formulas that are employed herein. This appendix is included for guidance only.

TABLE II. Standard English-Metric equivalents.

Volume:		
1 fluid ounce	29.57 milliliters	
1 pint	0.47 liter	
1 quart	0.95 liter	
1 gallon	3.79 liters	
1 cubic foot	0.03 cubic meter	
NOTE: To change cubic inches to cubic feet, divide by 1728.		
Weight:		
1 ounce (avoirdupois)	28.35 grams	
1 pound (avoirdupois)	453.59 grams	0.454 kilogram
2.2 pounds	1.00 kilogram	
Length:		
1 inch	2.54 centimeters	25.4 millimeters
1 foot	30.48 centimeters	0.305 meter
39.37 inches	1.00 meter	
3.28 feet	1.00 meter	
25.00 feet	7.625 meters	
50.00 feet	15.25 meters	
NOTE. 10 millimeters equals 1 centimeter.		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

TABLE III. MIL-HDBK-775 - English-Metric conversions.

Temperature Conversion:			
To change degrees Celsius (C.) to degrees Fahrenheit (F.), multiply temperature by 1.8 and add 32 degrees F.			
To change degrees Fahrenheit (F.) to degrees Celsius (C.), subtract 32 from the temperature and divide by 1.8.			
Degrees Fahrenheit		Degrees Celsius	
-20		-29	
0		-18	
32		0	
60		16	
68		20	
85		29	
100		38	
125		52	
150		66	
212		100	
Volume (Liquids):		**Weight:**	
Gallons	Liters	Ounces	Grams
1	3.79	1	28.35
5	18.95	10	283.5
50	189.5		
55	208.45	(1000 grams = 1 Kilogram)	
Volume (Solids):			
Cubic feet	Cubic meter	Cubic feet	Cubic meter
1.0	0.03	6.0	0.18
1.2	0.036	6.5	0.195
1.3	0.039	8.4	0.25
2.0	0.06	9.0	0.27
2.5	0.075	10.0	0.30
3.0	0.09	12.0	0.36
4.1	0.123	30.0	0.90
4.7	0.141	60.0	1.80
5.3	0.16	66.5	1.995

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

TABLE III. MIL-HDBK-775 English-Metric conversions – Continued.

Pounds	Kilograms	Pounds	Kilograms
.5	0.23	100	45.4
1	0.45	125	56.75
2.2	1.0	144	65.37
10	4.54	150	68.10
11	4.99	200	90.80
40	18.16	250	113.50
50	22.7	400	181.6
52	23.61	500	227.0
65	29.51	1000	454.0
75	34.05	1700	771.8
Inches	Centimeters	Inches	Centimeters
0.025	0.06	4.0	10.16
0.030	0.08	5.0	12.70
0.035	0.09	6.0	15.24
0.042	0.11	7.0	17.78
0.063 (1/16)	0.16	8.0	20.32
0.125 (1/8)	0.32	9.0	22.86
0.188 (3/16)	0.48	10.0	25.40
0.25 (1/4)	0.64	12.0	30.48
0.375 (3/8)	0.95	14.875	37.79
0.50 (1/2)	1.27	18.0	45.72
0.625 (5/8)	1.59	20.0	50.80
0.75 (3/4)	1.91	26.875	69.53
0.875 (7/8)	2.22	30.0	76.20
1.0	2.54	32.0	81.28
1.25	3.18	40.0	101.60
1.5	3.81	45.5	115.57
2.0	5.08	50.0	127.0
2.5	6.35	53.0	134.62
2.75	6.99	60.0	152.4
3.0	7.62	80.0	203.20

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army – SM
Navy – SA
Air Force – 69
DLA – DH

Preparing Activity:
Army – SM
(Project PACK-2021-009)

Review Activities:
Army – AT, AR, CR4, EA, MD
Navy – AS, OS
Air Force – 70, 71
DLA – CC

User Activities:
Army – MR, MT
Navy – BC, MC, SH, YD

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at **https://assist.dla.mil/** .

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:49Z
Check the source to verify that this is the current version before use.